

The two geographies of urban deprivation: everyday services and emergency care in 67 European cities

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Abstract

Critical infrastructures sustain urban life in two distinct ways: through the everyday services people reach by choice and through the emergency care that must reach them in minutes. Research has measured access to both as travel time, yet a uniform minutes scale says little about welfare. Twenty-five minutes instead of twenty to a pharmacy changes daily life marginally; the same five minutes on an ambulance route can be fatal. We therefore measure potential deprivation, transferring calibrated loss functions from humanitarian logistics [1]: a saturating deprivation *level* for walkable everyday services, anchored on the 15-minute-city threshold [2], and an escalating deprivation *cost* for emergency care, anchored on clinical response benchmarks, computed on a 100 m grid across 67 European functional urban areas in 24 countries. The two regimes co-evolve differently with city size. Size buys everyday access while emergency relief shows no detectable size gradient, and everyday inequality rises with size while emergency inequality never falls. Where the regimes diverge is structured by country and macro-region rather than by size, and the divergence culminates in a severity ordering of emergency-periphery coverage whose extreme is five capital-city emergency deserts that access averages render invisible. Compounding, the same residents deprived in both regimes, exceeds statistical independence almost everywhere, is strongest in Central-Eastern Europe, and falls on families with children in nearly nine of ten cities. Because everyday and crisis capability are distinct [3], planning for one regime cannot substitute for the other; deprivation analysis shows where each demands its own remedy.

Significance statement. Cities are widely evaluated by how fast residents reach services. This study shows that travel time alone conceals the welfare stakes of access. Measuring anchored deprivation instead of minutes across 67 European city regions reveals that growth improves daily-service access while leaving emergency coverage unchanged, that inequality in daily access widens with city size, and that five European capitals harbour emergency deserts invisible to conventional access statistics. The burden of combined everyday and emergency deprivation falls most consistently on children. For planners, everyday inequality emerges as a big-city problem and emergency deprivation as a national coverage problem, each requiring different instruments.

Keywords: accessibility; deprivation cost; 15-minute city; emergency medical services; urban scaling; spatial equity; resilience

1 Introduction

Urban critical infrastructure serves residents on two clocks. On the slow clock, people choose when to walk to a pharmacy, a supermarket, a school or a park; these trips are frequent, substitutable and forgiving of a few extra minutes. On the fast clock, an ambulance or an emergency department must reach a patient, and minutes translate into survival probabilities. A large literature measures the slow clock through proximity and accessibility metrics, most visibly in the 15-minute-city programme [2, 4] and in global minutes-based accessibility mapping [5, 6]. A parallel literature documents that access is unequally distributed within cities [7]. Both traditions share one measurement decision: access is a function of travel time alone, applied uniformly across services.

That decision discards welfare information. The burden of poor access is a function of what the service is for. For substitutable everyday services the burden saturates: beyond some threshold a household reorganises its routines, and additional minutes change little. For time-critical emergency care the burden escalates: every additional minute keeps costing, at an increasing rate. Humanitarian logistics has formalised this distinction as *deprivation*: the welfare cost of time spent without access to a good or service, estimable as a function of deprivation time [1, 8]. Deprivation *levels* for substitutable needs follow a saturating S-curve [9]; deprivation *costs* for critical needs are convex and unbounded, with discrete-choice and stated-preference estimates available for goods and for emergency ambulance services specifically [10–12]. This paper transfers those functional forms to intra-urban access and calibrates their curvature to domain anchors: the 15-minute-city threshold for the everyday regime, and the emergency-medical-service (EMS) response benchmarks for the emergency regime, where deprivation is negligible below the 3–4 minute ideal, accumulates through the widely used and empirically contested 8-minute urban response standard, and reaches the benchmark cost at the 10–15 minute upper target [13–17].

Measuring the two regimes separately is motivated by more than measurement hygiene. Urban resilience research defines resilience through equitable access to essential services [18, 19], finds that equality of access and network resilience move together [20], and shows empirically that everyday vulnerability profiles do not predict crisis-time capability [3]. If the everyday and emergency regimes are distinct capabilities, a city may deliver one and fail the other, and the population for whom both fail carries a compounded burden no single-regime statistic reveals.

We ask how the two deprivations co-evolve across the European urban hierarchy. The study covers 67 functional urban areas (FUAs) in 24 countries, from 101,050 to 12.9 million inhabitants, sampled by a stratified design over four macro-regions and four size strata, and analysed on a common 100 m population grid. Following the urban-scaling tradition we estimate log-log elasticities of city outcomes on population [21], reading trajectories from the cross-sectional size gradient under a harmonised city definition [22]. One qualifier applies to everything that follows and is stated here once: every size gradient in this paper is cross-sectional, and any phrase such as “as cities grow” means the space-for-time reading of that gradient, never a longitudinal observation.

Four research questions organise the analysis. RQ1: do larger cities provide better everyday services and emergency capability, in levels and in inequality? RQ2: where do the two deprivations diverge, and what structures the divergence: size, country, region, or coverage? RQ3: who carries compounding deprivation, in terms of co-location typology and vulnerability strata? RQ4: what does analysing deprivation add over access metrics?

The paper contributes, first, a two-regime deprivation framework, with functional forms transferred from the literature, curvature calibrated to anchors, substitution through a soft-minimum, congestion through a two-step floating catchment area (2SFCA) adjustment [23, 24], computed comparably across 67 FUAs. Second, evidence of regime-specific agglomeration: size buys everyday access and no detectable emergency relief, with the difference itself tested and the emergency null bounded by equivalence tests, while size increases everyday inequality.

Third, the geography of the divergence: country-level regional structure and a one-dimensional severity ordering of emergency-periphery coverage that replaces any city typology and whose extreme, five capital-city emergency deserts, is invisible to access averages. Fourth, an account of compounding and its carriers: co-location of the two deprivations above statistical independence nearly everywhere, with children carrying the burden consistently at the European level. Fifth, a robustness architecture usable as a template, combining rank-invariance by construction, a specification curve, rank-agreement statistics, a routing-engine cross-check and an access-based re-estimation of every headline.

2 Results

Two deprivation surfaces for 67 cities

For every populated 100m grid cell of each FUA we compute two indicators. The *everyday deprivation level* is a number between 0 and 1. It is built from the effective walk time to each of five everyday service categories (general practitioner, pharmacy, supermarket, school, green space), where “effective” means congestion-adjusted (a facility serving many people relative to its capacity counts as farther) and substitution-aware (several nearby options count as slightly closer than the nearest alone). Each service’s effective time passes through a saturating loss function whose inflection sits at the 15-minute-city threshold; the city-block indicator is the weighted mean over services. A value near 0 means all five services are comfortably walkable; a value near 1 means the cell is effectively without walkable everyday services, and the level no longer changes as times grow further.

The *emergency deprivation cost* is an open-ended positive number. It applies a convex cost function to the drive time to the nearest emergency facility (emergency department or ambulance station), calibrated so that cost is negligible below the 3–4 minute EMS ideal, reaches about a third of the benchmark cost at the contested 8-minute urban standard, and equals 1 at the 15-minute upper target; beyond 15 minutes it keeps escalating. All emergency costs in this paper are expressed in multiples of that 15-minute benchmark cost $g(15)$: a cell at 2.0 carries twice the deprivation cost of a cell whose ambulance arrives at the 15-minute benchmark.

The two regimes are deliberately different mathematical objects: a bounded level for substitutable services (once everyday access is bad enough, additional minutes change nothing, because routines adapt) and an unbounded cost for time-critical care (additional minutes always matter). Forcing both into levels would assert that a 60-minute ambulance is no worse than a 30-minute one; forcing both into costs would assert that living far from a supermarket is unboundedly catastrophic. The asymmetry is the two-regime claim itself, and it imposes a discipline used throughout: only unit-invariant statistics cross regimes, namely elasticities, Ginis (which are mean-normalised, the same move as comparing an income Gini with a wealth Gini) and rank-based measures. Raw magnitudes are never compared across regimes.

Table 1 summarises the sample; Table S2 in the SI lists every city. Fig. 1 gives the overview of all 67 cities at once: each city is a point in the plane spanned by its within-city inequality (Gini) of everyday deprivation and of emergency deprivation, with whiskers spanning the min–max envelope over the deprivation-curvature sensitivity grid, so a city’s position is shown at the precision the calibration actually supports.

Because all cross-city inference must respect that cities nest within countries, every scaling claim below cites a wild cluster bootstrap p-value with country clusters [25, 26], and every regional contrast cites a permutation test that reshuffles whole countries across macro-regions. City-level p-values are anti-conservative for such questions and are never cited as evidence.

Table 1: Sample composition and median city indicators by macro-region. 67 functional urban areas (Eurostat/OECD definition: city plus commuting zone), stratified draw over 4 macro-regions $\times$ 4 size strata (10^5 – $2.5 \cdot 10^5$, $2.5 \cdot 10^5$ – 10^6 , 10^6 – $5 \cdot 10^6$, $> 5 \cdot 10^6$ inhabitants; 15/18/30/4 cities), 113.9 million residents in total. $\bar{D}_{ev}$: population-weighted mean everyday deprivation level (0–1); $\bar{C}_{em}$: mean emergency deprivation cost (multiples of $g(15)$; not comparable to $\bar{D}_{ev}$ in magnitude); G : within-city population-weighted Gini; ρ : Spearman rank coupling of the two surfaces; HH: compounding population share at the median split. Source: `cities_descriptives.csv`.

Region	Cities	Countries	Pop. (M)	$\bar{D}_{ev}$	$\bar{C}_{em}$	G_{ev}	G_{em}	ρ	HH
North	13	4	11.3	0.41	1.81	0.39	0.47	0.47	0.34
West	16	5	37.1	0.23	0.74	0.52	0.41	0.36	0.32
South	18	4	39.8	0.29	1.07	0.42	0.42	0.38	0.32
CEE	20	11	25.6	0.30	1.48	0.51	0.54	0.55	0.36
All	67	24	113.9	0.29	1.14	0.49	0.44	0.42	0.33

2.1 City size buys everyday access; emergency relief shows no detectable size gradient

Mean everyday deprivation falls systematically with city size (Fig. 2). The log-log elasticity is -0.196 (country-clustered SE 0.023, $p_{wild} = 0.0002$, $R^2 = 0.44$): each e-fold ($\approx 2.7\times$) of population buys roughly 19% lower mean everyday deprivation. The estimate is stable under country fixed effects (-0.17), leave-one-out city deletion (envelope -0.204 to -0.185), robust estimators (Huber -0.196 , median regression -0.223) and exclusion of the emergency-desert group (-0.203).

Mean emergency deprivation cost shows no detectable size gradient: the elasticity is -0.059 (clustered $p = 0.16$, $p_{wild} = 0.21$). The null is bounded, and the bound is part of the result: two one-sided equivalence tests [27] place the emergency elasticity within ± 0.13 ; equivalence at the stricter ± 0.10 bound is not shown, so the claim is “no detectable scaling, bounded within ± 0.13 ”, never “proven flat”. The difference between the two slopes is itself significant: a paired regime test within cities puts the everyday slope 0.137 below the emergency slope ($p_{wild} = 0.002$).

Agglomeration benefits are therefore regime-specific. Bigger cities deliver everyday services closer to people; proximity to emergency care does not improve detectably with size (RQ1, levels).

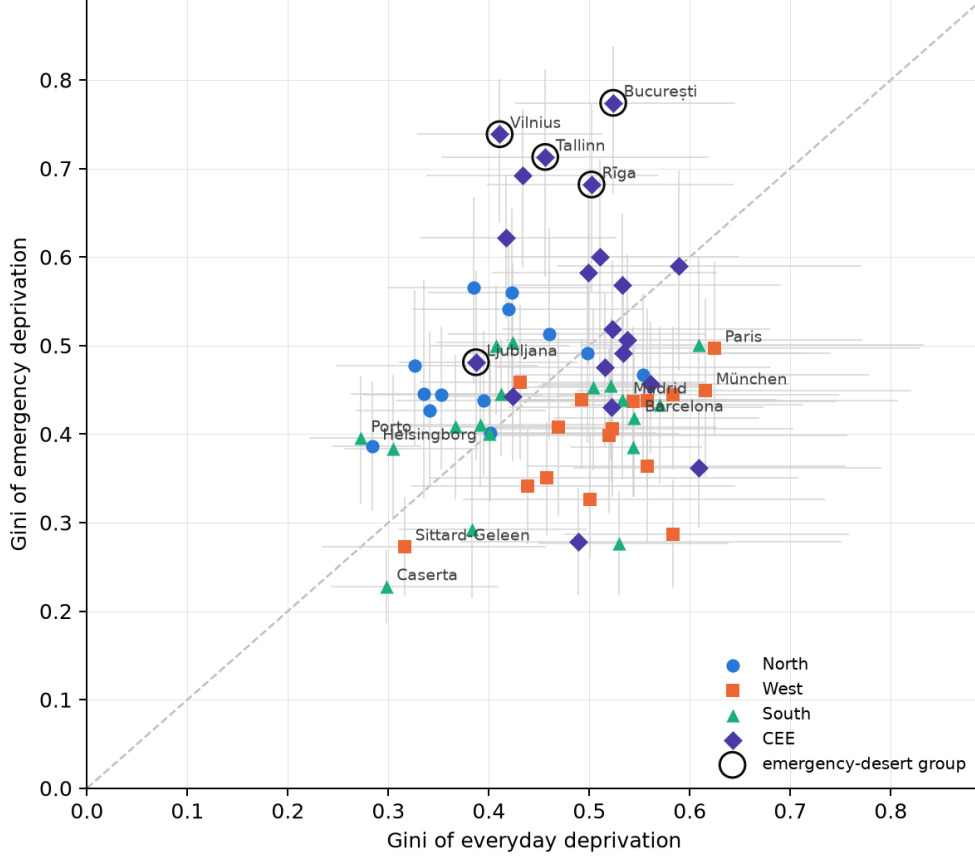
2.2 Inequality scales with size only in the everyday regime

Within-city inequality behaves differently again (RQ1, inequality). The Gini of everyday deprivation rises with population: elasticity $+0.062$ ($p_{wild} = 0.0012$). The emergency Gini is size-flat at the baseline parameterisation ($+0.004$, $p_{wild} = 0.84$), and the paired slope difference of -0.058 is significant over all 67 cities ($p_{wild} = 0.029$). Size therefore buys mean everyday access at the price of a wider spread around that mean, while emergency inequality is set elsewhere.

The everyday half of this claim is defended by a specification curve [28] rather than by any single calibration (Fig. 3): re-estimating the slope under all 14 deprivation parameterisations (the curvature grid and the functional-form swaps of Section 3.1) keeps it positive throughout ($+0.026$ to $+0.075$) and significant in 13 of 14. The exception is the concave Box-Cox everyday form swap ($+0.026$, $p = 0.083$), which we state rather than smooth over.

The emergency half is conditional, and we report it as a finding rather than resolve it by choice of parameterisation. The EMS benchmarks define the cost scale up to the 15-minute cut-off, and all three literature-grounded escalation forms are calibrated to coincide there. Beyond the cut-off no benchmark exists, and that is exactly where the forms diverge. Under the bounded survival-based curve (1 = full deprivation, saturating around 30–45 minutes) the emergency Gini is size-flat ($+0.001$, $p = 0.97$). Under polynomial Box-Cox escalation, the published baseline, it

Everyday-vs-emergency inequity plane (colour = macro-region; grey whisker = deprivation-curvature envelope)



67 cities · routing: r5 (friction = declared sensitivity variant) · facilities: OSM/Overpass, expiry-refreshed extractions · generated 2026-08-18 · cross-sectional space-for-time

Figure 1: All 67 cities in the inequality plane. Horizontal axis: within-city Gini of the everyday deprivation level; vertical axis: within-city Gini of the emergency deprivation cost. Whiskers span the min–max envelope of each Gini over the deprivation-curvature sensitivity grid (Section 3.1), so each city is placed at the precision the calibration supports. Colours mark the emergency-coverage severity grades of Section 2.3 (covered, partial desert, desert).

is size-flat across the whole curvature sweep (-0.006 to $+0.014$, none significant), in agreement with the deprivation-free travel-time Gini, itself size-flat (-0.015 , $p = 0.42$). Under exponential escalation the Gini rises with size ($+0.098$, $p = 0.0003$); the mechanism, verified in the data, is that exponential pricing turns the Gini into an index of extreme-tail depth (it saturates at a median of 0.96, decouples from the time distribution’s inequality, $\rho = 0.04$, and tracks the beyond-45-minute population share, $\rho = 0.45$), and how many minutes past the cut-off the worst-served residents sit does grow with size, even though the tail’s population share does not (beyond-30-minute share elasticity -0.003 , $p = 0.99$). Two statements hold under every parameterisation: everyday inequality rises with size, and emergency inequality never falls with size. Whether emergency inequality is flat or rising is decided by the beyond-benchmark escalation form, which no clinical benchmark pins down; it is flat under two of the three hypotheses and rising under the third.

2.3 The geography of divergence: country-level structure and a coverage gradient

If size does not set emergency deprivation, what does? The divergence between regimes has a geography, and the honest unit for testing it is the country: macro-region is a property of countries, so we permute whole countries across regions (Fig. 4). The emergency Gini differs

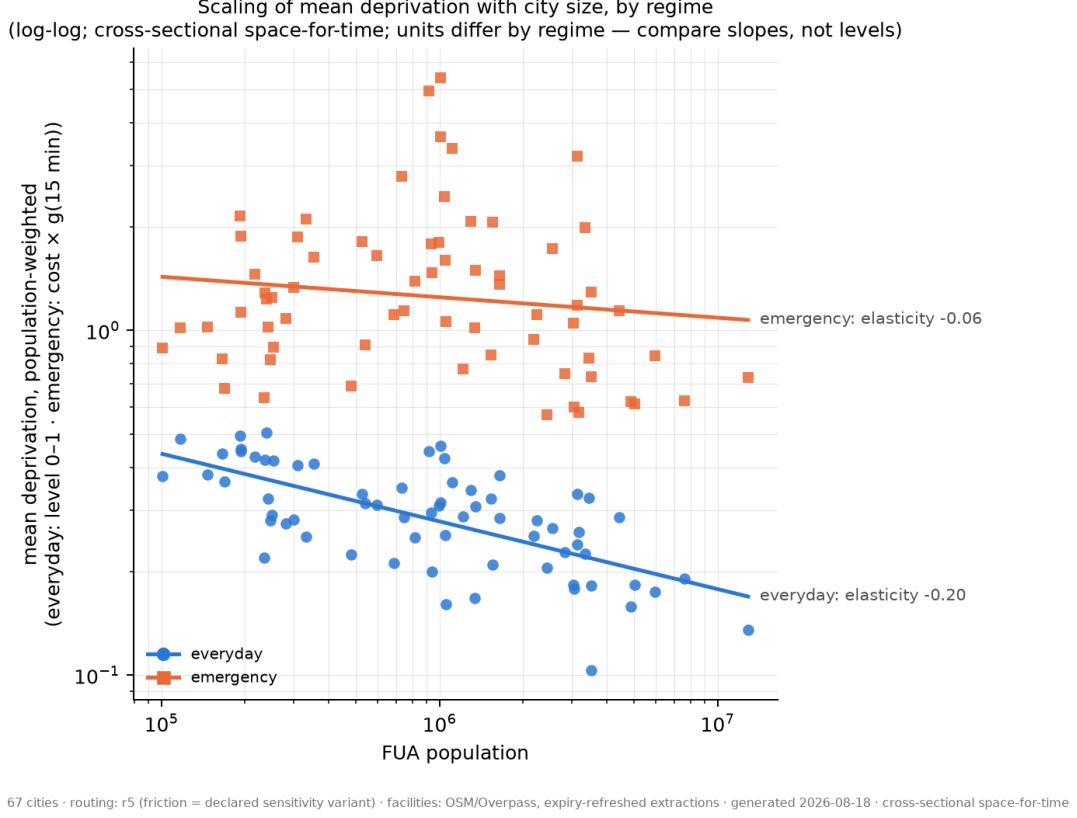


Figure 2: Regime-specific agglomeration. Log-log regressions of population-weighted mean deprivation on FUA population across the 67 cities. Everyday deprivation level (left): elasticity -0.196 , country-clustered wild bootstrap $p = 0.0002$, $R^2 = 0.44$. Emergency deprivation cost (right): elasticity -0.059 , wild $p = 0.21$, bounded within ± 0.13 by equivalence testing. The paired slope difference of $+0.137$ is significant at wild $p = 0.002$. Gradients are cross-sectional (space-for-time).

by macro-region (Cohen $f = 0.74$, $p_{\text{perm}} = 0.0024$), as do the divergence gap ($p_{\text{perm}} = 0.010$), the everyday–emergency coupling ρ ($p_{\text{perm}} = 0.0015$) and both compounding measures ($p_{\text{perm}} = 0.0008$ for the HH share, $p_{\text{perm}} = 0.0023$ for the continuous intensity). Central-Eastern Europe compounds: the region pairs the highest emergency inequality (median emergency Gini 0.54) with the strongest coupling between the two surfaces (median $\rho = 0.55$), so the same populations tend to carry both deprivations. The West’s inequality problem is everyday-side (median divergence gap -0.12); the North’s is emergency-side ($+0.09$). The everyday Gini’s apparent regional contrast does not survive country-level testing ($p_{\text{perm}} = 0.27$ despite a city-level ANOVA $p = 0.0007$), a cautionary illustration of why city-level inference over-claims here.

Do the 67 cities fall into types? No. k-means clustering on the full standardised city-vector separates 62 cities from 5 (silhouette 0.65, bootstrap ARI 0.87 [29, 30], Gaussian-null $p = 0.001$), and the isolated group is an outlier group rather than a typology: its agreement with the region partition is ARI = 0.001. Those five are Bucureşti, Rīga, Vilnius, Tallinn and Ljubljana, the emergency-desert capitals, separated by their beyond-threshold emergency travel-time tails. Ward clustering selects the same axis and cuts it more conservatively, isolating a nested subset of the two most extreme capitals (65/2). The nesting is evidence for a gradient: the two algorithms disagree about where to cut a severity ordering, and agree on the ordering itself. Re-clustering with the outliers removed finds one further split along the same axis at lower intensity: a 48/14 cut isolating partial-desert cities with real emergency tails (among them Oslo, Stockholm, Warszawa, Sofia, Zagreb, Palermo and Luxembourg; silhouette 0.38, bootstrap ARI 0.74, null $p = 0.001$, ARI vs region 0.07), inside which Ward again nests (56/6).

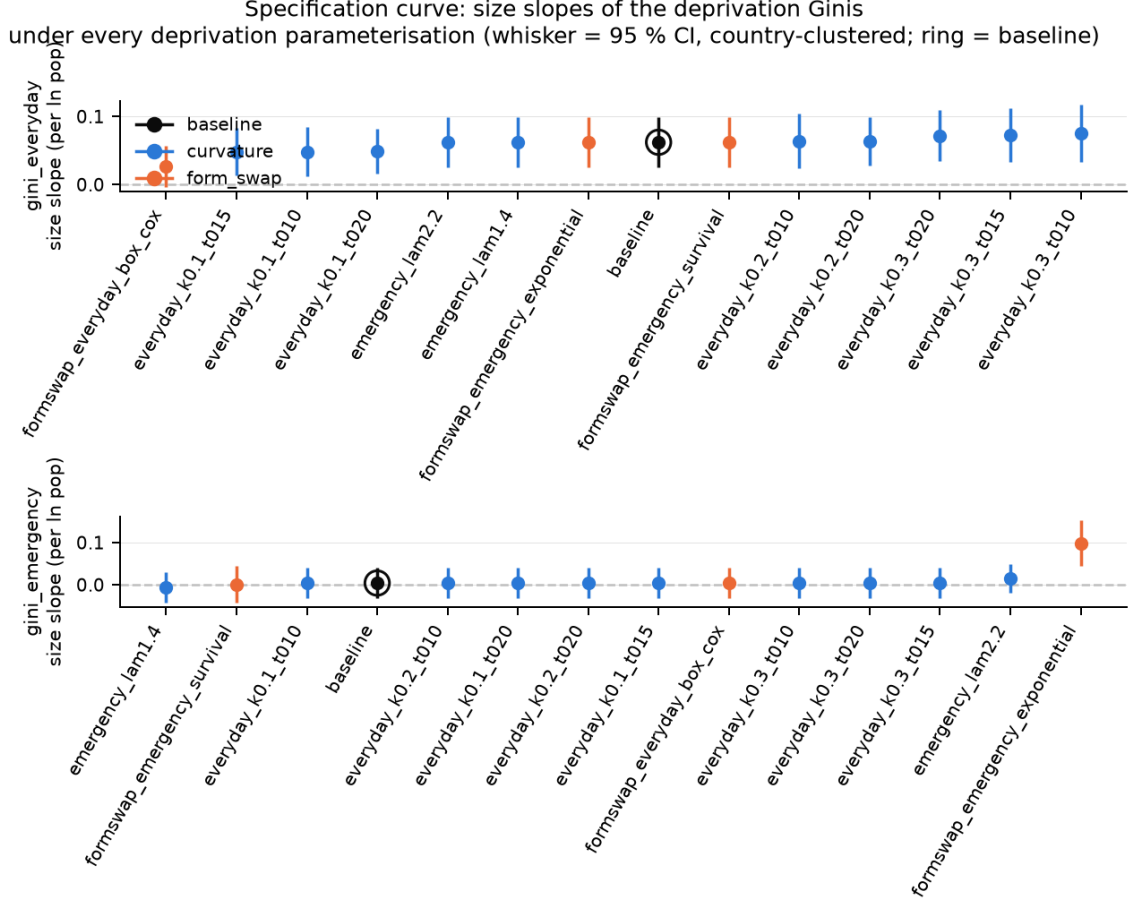


Figure 3: Specification curve of the inequality-scaling slopes. Each point is the size elasticity of a within-city Gini re-estimated under one of the 14 deprivation parameterisations (curvature grid; functional-form swaps). The everyday-Gini slope is positive under every specification (+0.026 to +0.075; significant in 13/14, the exception being the concave Box-Cox form swap at $p = 0.083$). The emergency-Gini slope splits by beyond-cut-off escalation form: flat under saturating and polynomial escalation, rising only under exponential escalation (+0.098, $p = 0.0003$).

A feature-set robustness check that re-runs the whole pipeline with the 8/15-minute benchmark shares added reproduces the main partition exactly (ARI 1.00) and the peeled cut at ARI 0.97, while flagging Brăila and Łomża, the two smallest CEE cities, as sitting on the covered/partial boundary; the same two cities sat on the covered side under the pre-refresh extraction state. The citable structure is therefore a one-dimensional *severity ordering* of emergency-periphery coverage, covered (48) → partial desert (14) → desert (5), with a soft boundary between covered and partial and a robust ordering; we refer to these as coverage grades and never as city types.

The coverage grade, and no other variable we test, sets the emergency level. In a single country-clustered model with $\ln$ population centred at the sample mean, the grade dummies shift mean emergency deprivation overwhelmingly (Wald $p \approx 1.3 \times 10^{-15}$); the median mean cost rises $1.02 \rightarrow 1.87 \rightarrow 3.37$ multiples of $g(15)$ across covered → partial → desert (Fig. 5). The slope-by-grade interaction is $p = 0.010$ under the current 48/14/5 grades but was $p = 0.17$ under the pre-refresh 50/12/5 grades; a two-city boundary change flips it, so we treat the interaction as boundary-sensitive and do not headline any grade-specific size gradient. The level statement carries the policy content: coverage class, not city size, sets emergency deprivation. Consistent with this, the divergence-gap measures show at most a weak size gradient (gap slope -0.066 per $\log_{10}$ population, $p = 0.012$, $R^2 = 0.06$; Fig. S2), so size is a poor predictor of where the two regimes part ways (RQ2).

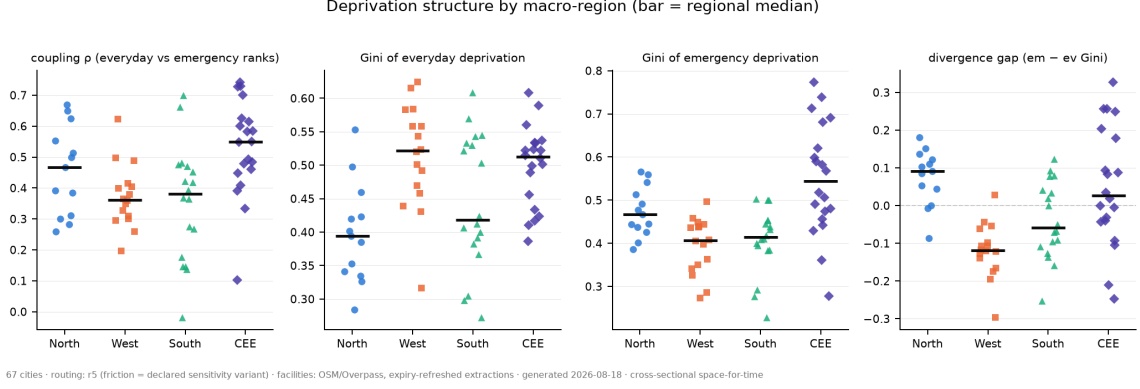


Figure 4: Regional structure of the divergence. Strip plots by macro-region for the indicators whose regional contrast survives country-permutation testing: emergency Gini ($p_{\text{perm}} = 0.0024$), divergence gap ($p_{\text{perm}} = 0.010$), coupling ρ ($p_{\text{perm}} = 0.0015$) and compounding (HH share $p_{\text{perm}} = 0.0008$; continuous intensity $p_{\text{perm}} = 0.0023$). The everyday Gini shows no regional contrast at the country level ($p_{\text{perm}} = 0.27$) and is not claimed. Region is a property of countries, so whole countries are permuted across regions; city-level p-values would be anti-conservative.

Table 2: The five emergency-desert capitals: access averages against deprivation. Median drive time to the nearest emergency facility and its ratio to the 67-city sample median (12 min) against mean emergency deprivation cost (multiples of $g(15)$) and its ratio to the sample value (1.14). Deprivation ratios of 2.45–4.72 coexist with median-time ratios of 0.71–1.92: the deserts are periphery-tail phenomena that access averages cannot see. Source: `desert_access_contrast.csv`.

City	Country	Median time (min)	Time ratio	Mean cost	Cost ratio
Vilnius	LT	14.0	1.17	5.39	4.72
Ljubljana	SI	23.0	1.92	4.96	4.34
Rīga	LV	13.5	1.13	3.37	2.95
București	RO	8.5	0.71	3.21	2.81
Tallinn	EE	13.0	1.08	2.80	2.45

The deserts are invisible to access averages. București’s median emergency drive time is 8.5 minutes, 0.71 times the sample median, which by any minutes-based average makes it look better-served than typical; its mean emergency deprivation cost is 2.81 times the sample value (Table 2). Across the five deserts, median-time ratios of 0.71–1.92 sit against deprivation ratios of 2.45–4.72. A well-served core pulls the median minute count down while a deep periphery tail carries extreme convex costs; only an anchored, convex cost function prices that tail (RQ4 preview, developed in Section 2.5).

2.4 Compounding is the norm, and children carry it

How often do the same residents lose on both clocks (RQ3)? At the median split of both percentile surfaces, the compounding (high-high, HH) population share would be 25 % if the two deprivations were independent. The observed HH share averages 33 % and exceeds independence in 66 of 67 cities; the rank coupling ρ between the two surfaces spans -0.02 to 0.74 (Fig. 6). The continuous compounding intensity, which removes the threshold’s leverage entirely (population-weighted mean of the smaller of the two percentile ranks; $1/3$ under independence, $1/2$ under perfect coupling), shows the same regional pattern ($p_{\text{perm}} = 0.0023$). Because the “how high is high” threshold moves the HH share roughly six times more than any accessibility-model assumption (Section 3.1), the continuous intensity is the headline compounding measure and every thresholded share travels with a threshold sweep. Fig. 7 shows the spatial anatomy across contrasting cities: compounding concentrates in peripheries where saturated everyday

Mean deprivation vs city size by emergency-coverage grade
(the k-means severity ordering: level is set by coverage class, not size; grade level shift $p = 1.3e-15$; slope-x-grade interaction $p = 0.010$ (country-clustered))

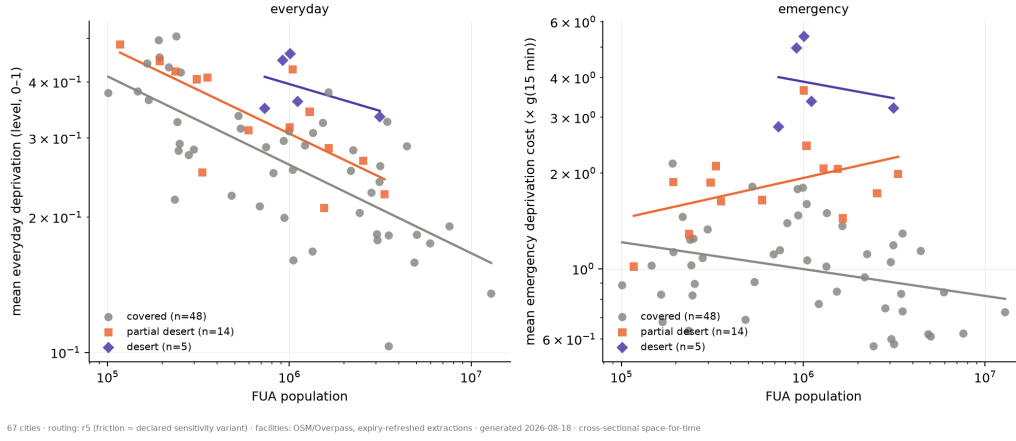


Figure 5: Emergency deprivation by coverage grade. Mean emergency deprivation cost (multiples of $g(15)$) against FUA population, coloured by the severity grades read from the two clustering passes (covered 48 / partial desert 14 / desert 5). The grade shifts the level decisively (country-clustered Wald $p \approx 1.3 \times 10^{-15}$; median levels 1.02 / 1.87 / 3.37), while the slope-by-grade interaction is boundary-sensitive and not headlined. Within-grade fits are plain OLS over few cities and shown for orientation only.

deprivation meets escalating emergency cost; the per-city maps for all 67 cities are in the SI gallery.

Who lives in the compounded areas? Pooling the census-harmonised age strata across all 67 cities against covered-cell references, children are over-exposed almost everywhere: cells in the top quartile of child share carry more everyday deprivation than their comparison cells in 88 % of cities (median ratio 1.27) and more compounding in 88 % (median HH-share gap +0.13), strongest in Central-Eastern Europe (median ratio 1.44) and the North (1.30) (Fig. 8). The elderly show no such pattern at the harmonised European level (median ratio 0.96, above parity in only 37 % of cities); sharper national-data results exist for single cities (in Hamburg, elderly 1.37, children 1.52, low-rent 1.61 against covered references) but use different definitions and are never pooled. The European claim is therefore about children, and the elderly result is a null at that level.

2.5 Deprivation is not access: what the framework adds

Re-running the size regressions on plain minutes answers RQ4 with three moves at once (Table 3). First, the headline claims survive dropping the deprivation layer entirely: the everyday gradient is present in median walk minutes (elasticity -0.237 , $p = 3.8 \times 10^{-5}$), and the emergency non-gradient is present in minutes too (-0.043 to -0.059 ; only the p90 tail reaches $p = 0.018$). No headline depends on the deprivation functions. Second, the deprivation outcome is better behaved: $R^2 = 0.44$ against 0.16 for median minutes, because the saturating loss discounts variation in the flat region of the curve that carries no welfare content. Third, the deserts of Table 2 are invisible in the minutes-based averages and exposed only by the anchored convex cost. The two Gini families agree closely in ranking (Pearson $r = 0.965$ everyday, 0.976 emergency), which is why the rank-based results are safe to call parameterisation-free while the level results are where the calibration carries information.

Table 3: Deprivation against pure access: size elasticities of deprivation-based and minutes-based outcomes (log-log, country-clustered SEs, 67 cities; cross-sectional space-for-time). The everyday gradient and the emergency non-gradient both survive in plain minutes; the deprivation outcome halves the unexplained variance on the everyday side. Source: `deprivation_vs_access.csv`.

Regime	Outcome	Elasticity	p	R^2
Everyday	mean deprivation level	-0.196	< 0.001	0.44
Everyday	median walk time	-0.237	< 0.001	0.16
Everyday	mean walk time	-0.178	< 0.001	0.31
Emergency	mean deprivation cost	-0.059	0.16	0.02
Emergency	median drive time	-0.043	0.09	0.04
Emergency	mean drive time	-0.043	0.05	0.04
Emergency	p90 drive time	-0.059	0.02	0.05

3 Discussion

3.1 How robust is the deprivation layer itself?

A paper that measures deprivation rather than access must expect its loss functions to be attacked first, and the answer has two halves that both need stating. Fig. 9 draws what is varied: the baseline everyday level function and emergency cost function; the Layer-1 curvature grid ($k \in \{0.1, 0.2, 0.3\}$ per minute $\times t_0 \in \{10, 15, 20\}$ minutes for the everyday logistic; $\lambda \in \{1.4, 1.8, 2.2\}$ for the emergency Box-Cox); the Layer-2 form swaps calibrated to the same domain anchors (a concave Box-Cox everyday level; exponential and bounded survival-based emergency escalations); and the comparison the whole framing rests on, the linear loss that a pure-access minutes average implicitly applies, matched to the baseline at 45 minutes. The gap between that straight line and the calibrated curves is the methodological claim in one picture: near the clinical threshold the cost function prices a marginal minute far more heavily than any average can, and past the 15-minute anchor the saturated level function prices it far less.

How far do results move under those variations (Table 4)? Levels are not robust, and are not meant to be: curvature alone moves a typical city’s everyday Gini by 0.24 Gini points (median envelope width across the 67 cities), which is exactly why the calibration carries information and why every slope claim is defended by the specification curve rather than by one parameterisation. Ranks and classes are robust: the same curvature grid moves the compounding share by 1.4 percentage points at the median, the form swaps by 0.9, against 29.8 for the “how high is high” threshold, a roughly 21-fold dominance of the threshold over the calibration; hence the continuous intensity as headline. City-level Gini orderings survive with two scoped exceptions: rank agreement with the baseline stays at $\rho \geq 0.90$ over the curvature grid and the everyday form swap, while both emergency escalation swaps reorder the emergency-Gini ranking (exponential $\rho = 0.19$; bounded survival $\rho = 0.49$) precisely because each changes what that Gini measures, a change of measurand reported as such rather than noise.

The wider robustness inventory reads as one block. Deprivation-variant flip cells, the population whose typology class changes under any deprivation parameterisation, average 18.1 % of city population (range 5.2–53.0 %), with the HH class share moving 2.1 percentage points on average and 6.1 at most. The clustering survives its own feature-set choice (main partition ARI 1.00, peeled 0.97 with benchmark shares included). The routing-engine cross-check bounds the engine’s leverage: switching engines moves travel-time levels by 34–314 % and flips the class of roughly 24 % of population per city, while aggregate class shares move at most 0.7 percentage points and the divergence ρ by around 10 %; per-cell map readings are therefore never findings, aggregate shares are. OSM completeness was benchmarked against national registries where available (Germany: pharmacies 0.976, hospitals 1.24) and the external INKAR benchmark passed. Fig. S1 assembles this architecture in one view. All of this supports one

Table 4: Sensitivity envelopes of the deprivation layer: median min–max envelope width across the 67 cities, per sweep axis and target. Curvature moves the Ginis (the calibration carries information); no axis of the calibration moves the compounding share materially compared with the “how high is high” threshold, which dominates by a factor of about 21. The form-swap emergency-Gini envelope (0.649) spans the three beyond-cut-off escalation hypotheses and is a boundary-of-validity statement, discussed with H2’s conditionality in Section 2.2. Source: `deprivation_sensitivity_summary.csv`.

Sweep axis	Everyday Gini	Emergency Gini	HH share
Curvature (Layer 1)	0.244	0.170	0.014
Form swap (Layer 2)	0.035	0.649	0.009
“How high is high” threshold	–	–	0.298

reading discipline used throughout the paper: rank-based results are near-invariant to the deprivation parameterisation, and level and inequality results are where the calibration carries the information that access measures cannot.

3.2 Implications for theory and practice

For resilience theory, the results give empirical content to the claim that everyday functioning and crisis capability are distinct capabilities [3, 19]. If resilience is defined as equitable access to essential services [18], then the two regimes measure two different resiliences, and they respond to different levers: the everyday regime to urban form and density, which agglomeration improves; the emergency regime to national emergency-system geography, which it does not. The compounding result sharpens the equality-resilience interplay documented for population-facility networks [20]: the pathological case is precisely when the same residents sit in the periphery of both systems, and we find that case above statistical independence in 66 of 67 cities, strongest where emergency inequality is highest. A city’s resilience portfolio cannot be read from either surface alone.

For urban analytics, the results argue that the measurement layer between travel time and welfare deserves the same scrutiny as the routing itself. The anchored loss functions change no rank-based conclusion, which is reassuring for the access literature; what they change is the outcomes’ statistical behaviour (halving unexplained variance on the everyday side) and the visibility of the tail phenomena that matter most (the deserts). The framework travels: forms are transferred from an estimated literature, curvature is calibrated to domain anchors that other domains possess in comparable form, and the robustness architecture (specification curve, rank agreement, engine cross-check, access re-estimation) is a template any anchored-measurement study can reuse.

For policy, the two regimes point at two different jurisdictions. Everyday inequality is a big-city problem: it rises with size under every parameterisation tested, so the cities gaining everyday access on average are the ones whose internal spread widens, and intra-urban service planning is the lever. Emergency deprivation is a national coverage problem: it is set by coverage grade rather than size, differs by country and macro-region, and its extreme cases are five capital cities whose peripheries lie beyond any urban administration’s reach alone. The severity ordering (covered, partial desert, desert) gives national emergency-system planners a screening variable that access averages do not provide. And the consistent carriers of compounded deprivation at the European level are families with children, in 88 % of the cities studied, which places the burden on a group whose daily radius is short and whose emergency risk is real.

3.3 Limitations

Every size gradient is cross-sectional; the space-for-time reading was flagged once and applies throughout. The analysis covers walking for everyday services and driving for emergency care;

public transit is deliberately out of scope, and the everyday regime is walk-only by design. Census strata arrive on a 1 km grid broadcast to the 100 m analysis grid, so they are neighbourhood attributes; employment is unreported in the census grid for Germany and France, so no harmonised SES slope exists there by design. Per-cell typology classes carry a roughly 24 % routing-engine flip share and are read only as spatial patterns. Population shares beyond fixed time thresholds contain the finite-fill constant assigned to service-deprived cells and are footnoted wherever used. The emergency-side external benchmark (INKAR) is unavailable, and the German registry counts carry a verify-before-publication flag.

Two OSM completeness specifics deserve statement rather than smoothing. Mapped GP density varies about 12-fold across countries in a way that reflects tagging conventions rather than health systems: the median FUA has 10.8 mapped GPs per 100k inhabitants, yet the country medians are 2.1 in Sweden, 2.3 in Portugal, 2.6 in Finland and 3.5 in Italy against 24.5 across the West, because primary care there sits in health centres OSM tags differently. Everyday deprivation levels are correspondingly overstated in those countries, so cross-country everyday *level* comparisons are not supported (one plausible reason the everyday-Gini regional contrast fails country-level testing). The size gradients are unaffected: mapped GP density is uncorrelated with city size (log-log slope +0.09, country-clustered $p = 0.29$), and every scaling claim is country-clustered. Separately, eight cities have only one of the two emergency services mapped (Athina, Brăila, Helsinki, Lahti, Norrköping, Szeged, Talavera de la Reina, Žilina); the composite renormalises weights over the services present, their median mean emergency cost is 0.197 against 0.169 for the rest, the expected direction at a modest size, and none of them is in the desert group, so no headline turns on it. Table S2 reports the count per city.

4 Materials and Methods

Design and sample. Cross-sectional multi-city design with one harmonised city definition (Eurostat/OECD functional urban areas) and trajectories read from the cross-sectional size gradient [21, 22]; no longitudinal component exists, and all outputs are labelled space-for-time. The sample is a stratified draw over 4 macro-regions $\times$ 4 population strata: 67 FUAs, 24 countries, 113.9 million residents, 101,050 to 12.9 million inhabitants (Table 1; Table S2).

Grid, facilities, routing. The analysis grid is the GHS-POP 100 m population grid [31], cells clipped to each FUA. Facilities come from OpenStreetMap via Overpass queries with expiry-refreshed extractions; Table 5 summarises the routed facility sets. Travel times are computed with the R5 routing engine via r5py [32] over the OSM street network: walking for the everyday regime, driving for the emergency regime, with emergency matrices reverse-routed (station-to-cell, the substantively correct direction for ambulance response; measured car asymmetry median 1 minute). A friction-surface engine [6] is retained as a declared sensitivity variant after a two-city cross-check showed city-specific, uncorrectable level bias (E.1).

Everyday deprivation (a level). Per service, travel time is congestion-adjusted by a two-step floating catchment area factor [23, 24]: facilities serving many people relative to capacity inflate the effective time of their users. A soft-minimum over reachable facilities then rewards substitutability (several similar options count as slightly closer than the nearest alone). The effective time passes through a logistic deprivation level function [9] with ceiling 1, inflection $t_0 = 15$ minutes (the 15-minute-city threshold [2]) and steepness $k = 0.2$ per minute; the city surface is the weighted mean over the five service categories (school and green-space sub-types weigh 0.5 each). Cells with no network path to any facility are masked; routable cells beyond a service’s cutoff receive a large finite time so they stay on the map at near-maximal deprivation.

Table 5: Facilities routed to (medians across the 67 FUAs) and population-weighted median travel times to the nearest facilities of each service. Everyday services are reached on foot, emergency services by car. Source: `accessibility_by_service_pooled.csv`.

Regime	Service	Facilities per FUA	Median time (min)
Everyday	general practitioner	93	7.9
Everyday	pharmacy	228	4.9
Everyday	supermarket	325	4.7
Everyday	school	409	4.4
Everyday	green space	636	3.8
Emergency	emergency department	6	12.0
Emergency	ambulance station	6	12.0

Emergency deprivation (a cost). The emergency surface applies a convex Box-Cox deprivation cost function [10, 12] with $\lambda = 1.8$ (transferred; solving λ from the anchors gives 1.89–1.95, inside the sensitivity sweep) to the drive time to the nearest emergency facility. The anchors are the EMS response benchmarks: cost negligible below the 3–4 minute ideal [13, 16], about a third of the benchmark cost at the 8-minute urban standard (contested: no survival effect for most patients in [13], supported in [14]; attributed via [17], whose WHO 2020 reference is an unpinnable URL), and 1 at the 15-minute upper target [15, 17]; convexity is supported independently by distance-mortality evidence [33]. The 8-minute mark cannot carry a half-max anchor: convexity caps $g(8)/g(15)$ at 0.427 given $g(4)/g(15) = 0.10$, so the two regimes’ anchor structures are genuinely different. All costs are reported in multiples of $g(15)$. Free-flow civilian drive times anchor the cost scale and are not literal response-time predictions (dispatch and turnout are unmodelled; lights-and-siren travel is faster than civilian flow; free-flow understates congested cores, which is conservative for the periphery findings).

Cross-regime standardisation. The everyday level and the emergency cost are on incomparable scales by design. Only unit-invariant statistics cross regimes: elasticities, Ginis and rank-based measures. The co-location typology is built on population-weighted mid-rank percentile surfaces; at the median split the 2×2 class table has one free number, the HH share, and statistical independence corresponds to $HH = 25\%$. The continuous compounding intensity is the population-weighted mean of the smaller of the two percentile ranks (1/3 independence, 1/2 perfect coupling, 1/4 perfect divergence). The divergence gap (emergency Gini minus everyday Gini) is the one cross-regime subtraction, a difference of dimensionless indices, form-dependent, and never a standalone headline.

Equity strata. Census-2021 1km strata (broad age groups) are broadcast to the grid and pooled across cities against covered-cell references (cells where the stratum’s variable is published); national fine-grid strata (rent, tenure) are per-city depth and never pooled. Inequality within strata uses the concentration index [34].

Cross-city inference. Cities nest in 24 countries, so scaling regressions report country-clustered standard errors with wild cluster bootstrap p-values (null imposed) [25, 26]; regional contrasts are tested by permuting whole countries across macro-regions, with a country-random-intercept mixed model as triangulation; the emergency null is bounded by TOST equivalence tests [27]; slope robustness uses a specification curve over all 14 deprivation parameterisations [28]; clustering diagnostics report silhouette [30], bootstrap and cross-partition adjusted Rand indices [29] and a Gaussian silhouette null. Every headline elasticity carries leave-one-out, Cook’s-distance, Huber and median-regression diagnostics.

Robustness harness. A structured (non-probabilistic) sweep recomputes rank-based targets over a defensible envelope: Layer 1 varies curvature within fixed forms; Layer 2 swaps the forms while pinning the same anchors (concave Box-Cox everyday; exponential and bounded survival-based emergency escalations); Layer 3 sweeps the accessibility model (soft-min κ , congestion exponent γ , bandwidth, k -nearest, mode set, unreachable policy); a threshold axis sweeps the typology cut. Results are reported as rank agreement, class-share envelopes, flip-cell shares and the specification curve (Section 3.1; Fig. S1).

Reproducibility. The pipeline is config-driven with provenance sidecars on every derived matrix; all tables and figures in this paper trace to the persisted 67-city state (every quoted number is checked against its source table by an automated audit of 133 checks run in CI). Code, configuration and the full result tables are available in the study repository.

Data availability. Population grid: GHS-POP R2023A [31]. Facilities and street networks: OpenStreetMap. Census strata: Eurostat Census 2021 grid. Per-city and cross-city result tables, per-city maps and sensitivity sweeps are persisted on the study repository’s results branch.

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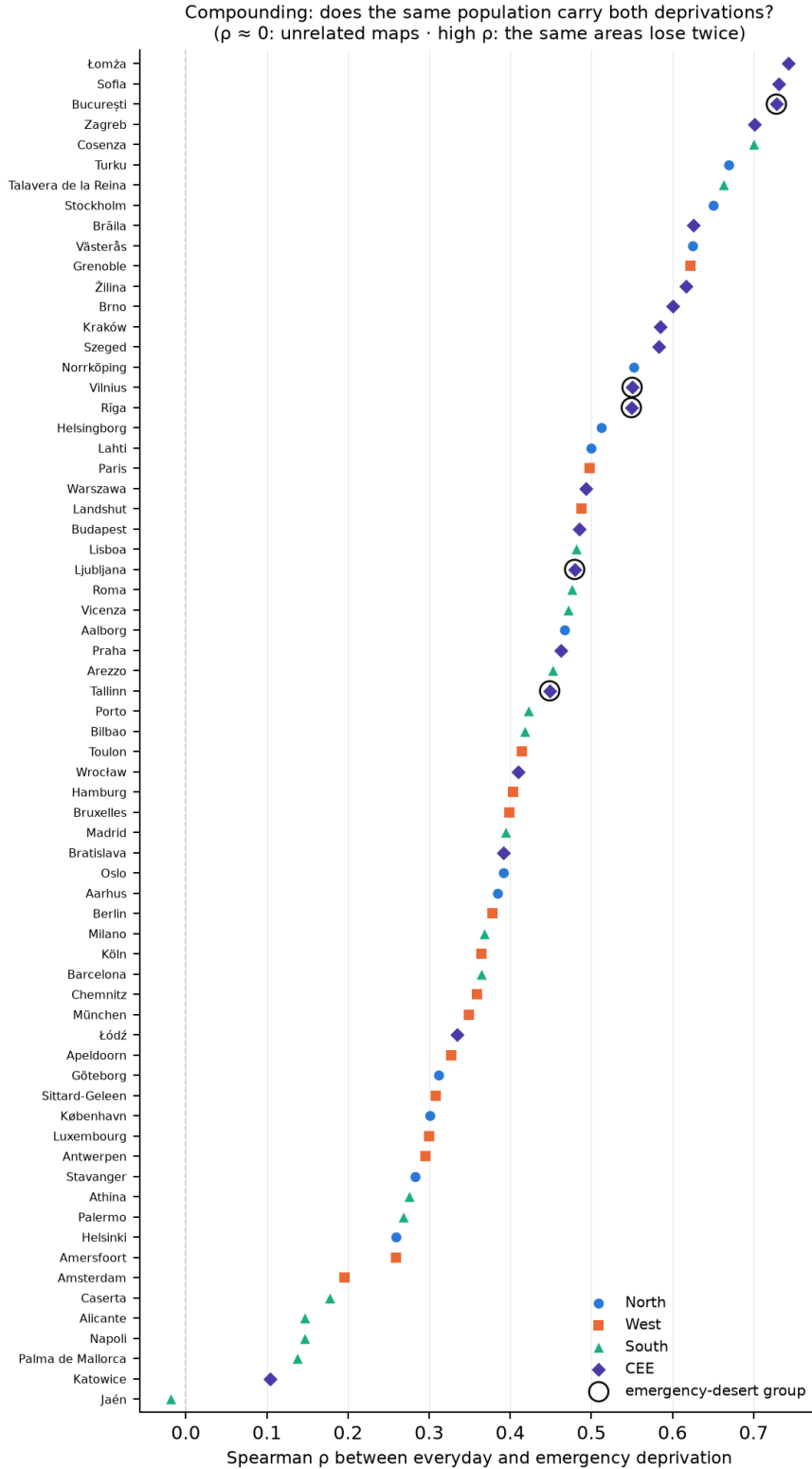
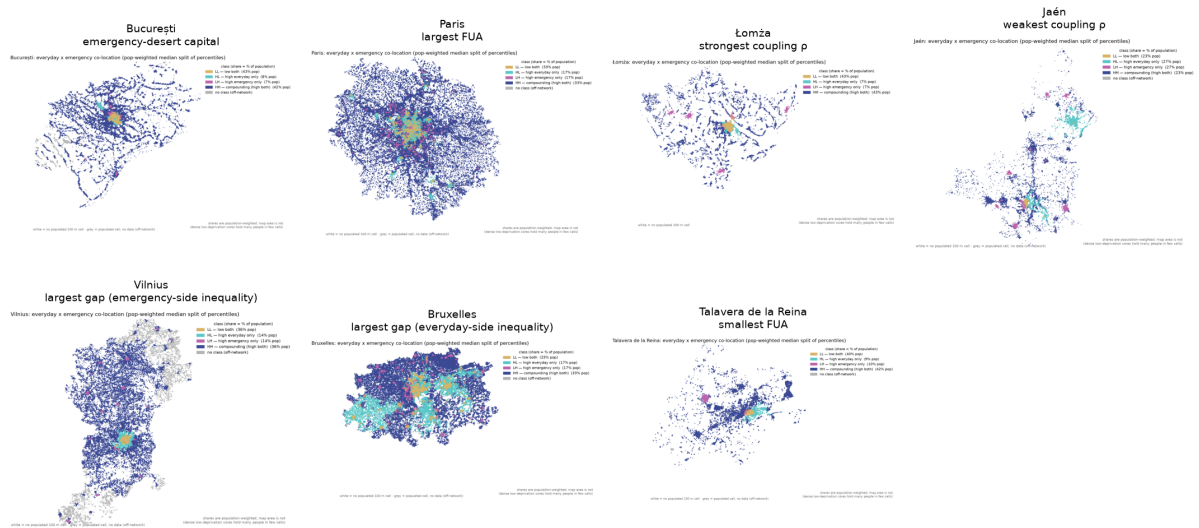


Figure 6: Coupling of the two deprivation surfaces across the 67 cities. Spearman ρ between the everyday and emergency population-weighted percentile surfaces, cities ranked; ρ spans -0.02 to 0.74 . The HH compounding share exceeds the 25% independence benchmark in 66 of 67 cities (sample mean 33%). Bars are annotated with each city's accessibility-model envelope where available; rank-based measures are invariant to the deprivation parameterisation.

Compounding deprivation, contrasting cities (everyday x emergency co-location at the median split)
Per-cell class assignment is engine-sensitive (~24 % flip, E.1): read the spatial patterns, not individual cells.



67 cities · routing: r5 (friction = declared sensitivity variant) · facilities: OSM/Overpass, expiry-refreshed extractions · generated 2026-08-18 · cross-sectional space-for-time

Figure 7: Compounding gallery: median-split co-location typology for a cross-section of cities (all 67 maps in the SI). Colours: LL both low, HL high everyday only, LH high emergency only, HH compounding. Shares in the legends are population-weighted; the maps are area-weighted, so a sparse periphery dominates visually at a modest population share. Maps are read for spatial pattern only: roughly 24 % of population flips class between routing engines, so no per-cell class is a finding.

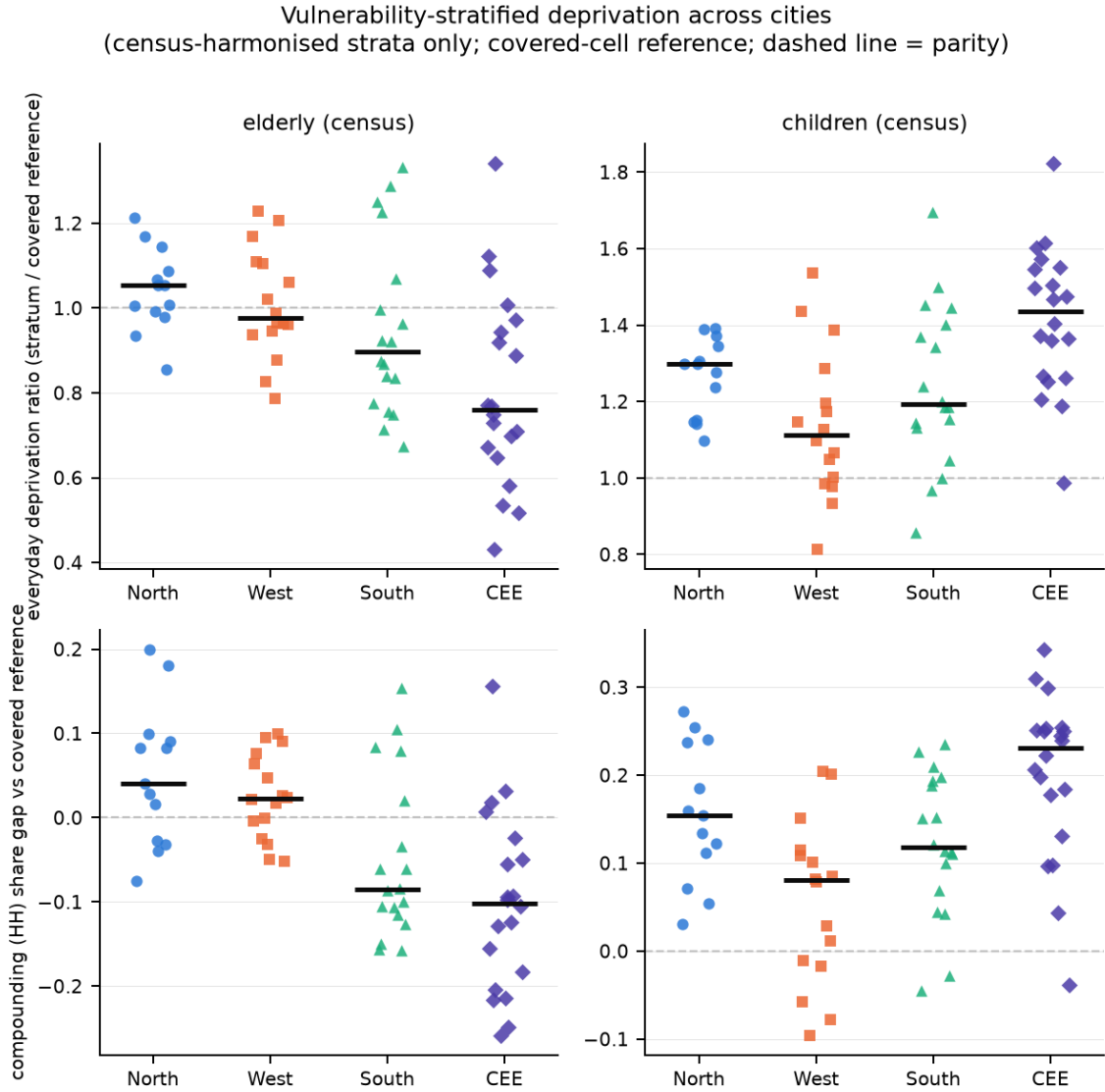


Figure 8: Who carries the deprivation: census-harmonised vulnerability strata pooled across the 67 cities (covered-cell references). Children (top-quartile child-share cells) experience higher everyday deprivation in 88 % of cities (median ratio 1.27) and higher compounding in 88 % (median HH gap +0.13), strongest in CEE (1.44) and the North. The elderly stratum sits at parity (median 0.96, above 1 in 37 % of cities); no European elderly claim is made. National fine-grid strata are per-city depth and are never pooled with these.

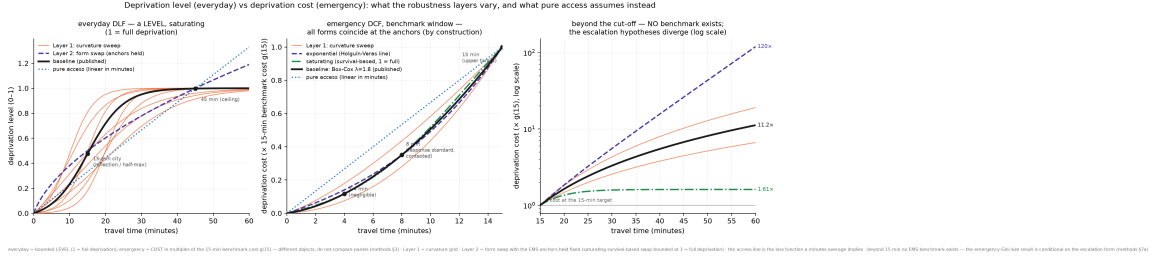


Figure 9: The deprivation functions and everything they are varied against. Left: everyday deprivation level (logistic, inflection at the 15-minute anchor; Layer-1 curvature sweep; concave Box-Cox form swap dashed), on its own 0–1 scale. Centre: emergency deprivation cost (convex Box-Cox, $\lambda = 1.8$, in multiples of $g(15)$) inside the benchmark window, where all forms coincide at the EMS anchors by construction (negligible at 4 min; ≈ 0.35 at the contested 8-min standard; 1 at the 15-min target). Right: beyond the 15-minute cut-off no benchmark exists and the three literature-grounded escalation hypotheses diverge (at 60 min: $1.61\times$ saturating, $11.2\times$ polynomial baseline, $120\times$ exponential; log scale). The dotted line in each panel is the linear loss a pure-access minutes average implies. Each panel stays on its own reporting scale; panels are never compared in magnitude.

Supplementary Information

The two geographies of urban deprivation

S1 The robustness architecture in one view

The robustness architecture: levels carry the calibration; ranks, classes and headlines do not depend on it
(rank agreement · sensitivity envelopes · access re-estimation · flip-cell vs class-share stability; 67 cities, all from docs/paper-pack/data)

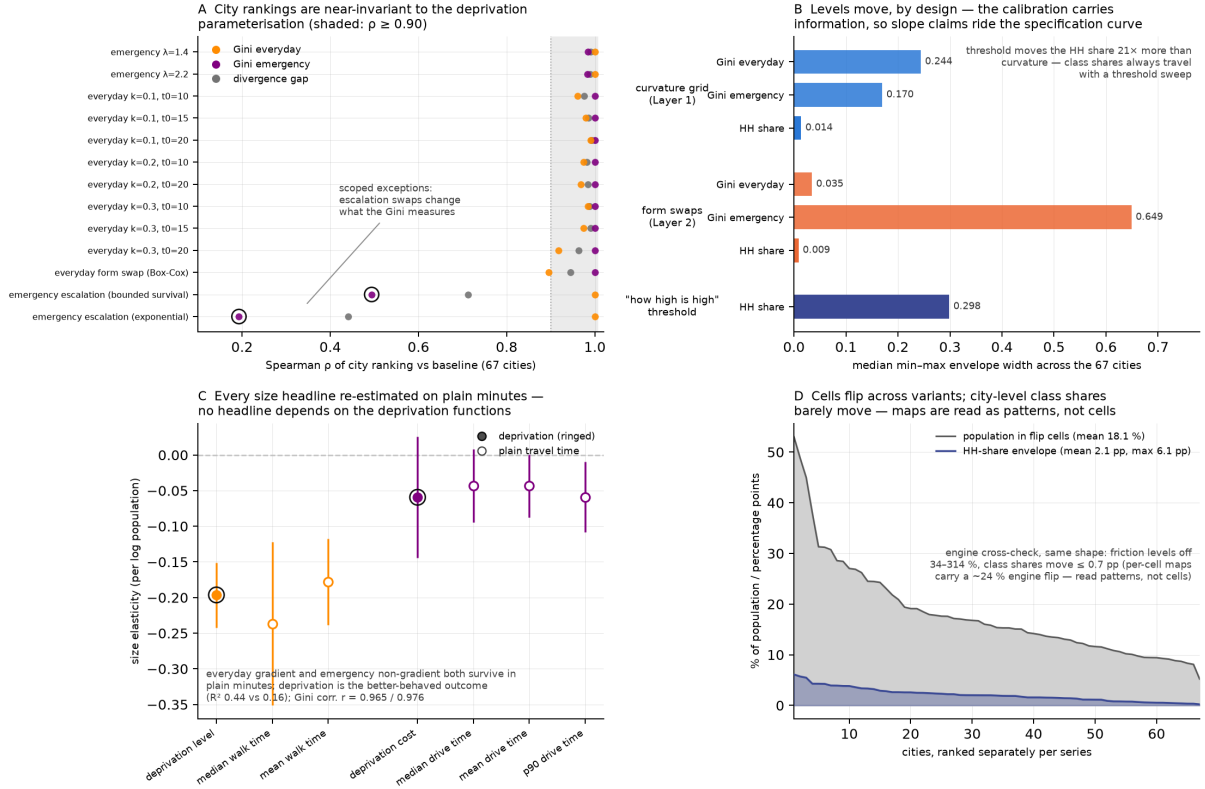
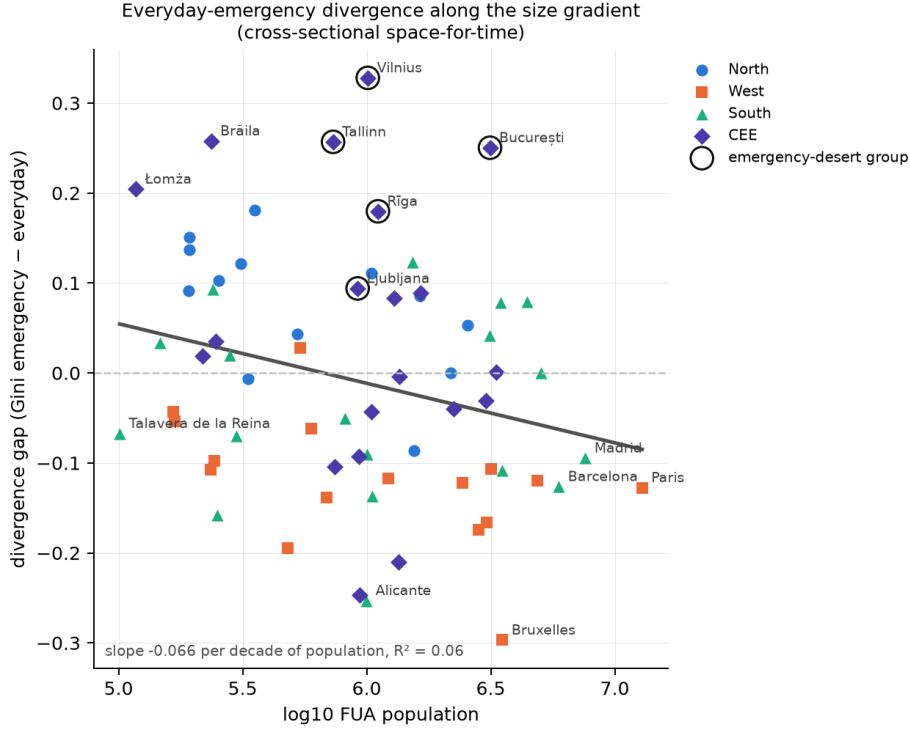


Figure S1: The robustness architecture (contribution claim 5). One view of the checks defending the results: rank agreement of city orderings across deprivation variants (min $\rho = 0.90$ over the curvature grid and everyday form swap; the two emergency escalation swaps are scoped exceptions that change the measurand, $\rho = 0.19$ exponential, 0.49 bounded survival); sensitivity envelope widths per axis (Table 4); access-based re-estimation of every headline (Table 3); and flip-cell against class-share stability (population-weighted flip share 18.1% on average, 5.2 – 53.0% across cities, against HH class-share movement of 2.1 pp average, 6.1 pp maximum), with the routing-engine cross-check quoted alongside (engine flip $\approx 24\%$, aggregate class shares ≤ 0.7 pp).

Rank-based results are near-invariant to the deprivation parameterisation; level and inequality results move with the calibration, which is where the calibration carries information. The flagged exceptions are exactly the two emergency escalation swaps, each of which changes what the emergency Gini measures (an extreme-tail-depth ordering under exponential escalation; a benchmark-window inequality ordering under the bounded survival curve), and they are reported as changes of measurand.



67 cities · routing: r5 (friction = declared sensitivity variant) · facilities: OSM/Overpass, expiry-refreshed extractions · generated 2026-08-18 · cross-sectional space-for-time

Figure S2: Size gradients of the divergence measures. The divergence gap falls weakly with size (-0.066 per $\log_{10}$ population, $p = 0.012$, $R^2 = 0.06$); the coupling ρ and the compounding share show no significant gradient ($p = 0.14$ and $p = 0.06$). Size is a poor predictor of where the two regimes diverge; the structure is regional and coverage-graded (main text Section 2.3).

S2 Size gradient of the divergence measures

S3 Validation exercises

E.1 Routing-engine cross-check. Re-running full cities under the friction-surface engine against the R5 baseline (Hamburg, Köln) shows travel-time levels understated by 34–314 % with city-specific Gini bias (emergency Gini -30% in Hamburg, -18.5% in Köln), which demoted friction to sensitivity variant. Aggregate typology class shares move ≤ 0.7 pp and the divergence ρ by $\leq 11\%$, while roughly 24 % of population flips per-cell class; hence per-cell map readings are never findings.

E.2 OSM completeness. Country-level benchmark against national registries where available: Germany, pharmacies ratio 0.976, hospitals 1.24 (the hospital ratio carries a verify-before-publication flag on the registry side). The per-city facility counts (`accessibility_by_service_cities.csv`) provide the cross-country completeness picture the single-country registry benchmark cannot, and surface the GP tagging artefact discussed in the limitations.

E.3 External benchmark. The German INKAR accessibility indicators reproduce the study’s German pharmacy and supermarket access patterns (passed); no comparable emergency-side benchmark is available.

E.4 Resolution and E.5 face validation. Quantile-curve comparisons of regime times under both engines and per-anchor-city face validation documents are maintained in the study repository.

Re-anchor verification. The emergency reporting anchor moved from $g(45)$ to $g(15)$ in August 2026. For the three cities re-seeded from stored artifacts the verification is exact: every rank-based indicator is identical to machine precision ($\max |\Delta| \approx 10^{-16}$) and the level ratio equals $g(45)/g(15) = 6.7307$ to 10^{-15} ; the other 64 cities were byte-identical between the two runs.

S4 Indicator glossary

Table S1: Compact glossary of the indicators used in this paper.

Term	Meaning
FUA	Functional urban area (Eurostat/OECD): city plus commuting zone; the unit “city”.
Everyday regime	Five walkable service categories (GP, pharmacy, supermarket, school, green space); deprivation is a bounded <i>level</i> in $[0, 1]$.
Emergency regime	ED hospital and ambulance station, reached by car; deprivation is an unbounded convex <i>cost</i> in multiples of $g(15)$.
$\bar{D}_{\text{ev}}, \bar{C}_{\text{em}}$	Population-weighted mean deprivation per regime; scales are incomparable across regimes.
$G_{\text{ev}}, G_{\text{em}}$	Population-weighted within-city Gini of each surface.
Divergence gap	$G_{\text{em}} - G_{\text{ev}}$; positive = emergency-side inequality problem.
ρ	Spearman correlation of the two population-weighted percentile surfaces.
LL/LH/HL/HH	Median-split co-location classes; HH = compounding. At the median split the table has one free number (HH); independence = 25 %.
Compounding intensity	Population-weighted mean of min(everyday pct, emergency pct); 1/3 independence, 1/2 coupled, 1/4 divergent.
Coverage grade	Severity ordering from the two clustering passes: covered / partial desert / desert.
Elasticity	Slope of log outcome on log population (space-for-time).
Wild cluster bootstrap	Country-clustered resampling giving reliable p-values with 24 clusters; the citable p for scaling claims.
p_{perm}	Permutation p-value reshuffling whole countries across macro-regions; the primary regional test.
TOST	Two one-sided equivalence tests; positive evidence a slope lies within a bound.
Specification curve	Headline regression re-estimated under every deprivation parameterisation.
Flip cells	Cells whose typology class changes under any deprivation variant.

S5 Per-city table

Table S2: The 67 functional urban areas, largest first. Grade: emergency-periphery coverage grade (Section 2.3); n_{em} : number of mapped emergency services (2 = both ED hospital and ambulance station); $\bar{D}_{ev}$: population-weighted mean everyday deprivation level (0–1); $\bar{C}_{em}$: mean emergency deprivation cost in multiples of the 15-min benchmark cost $g(15)$; G_{ev} : population-weighted Gini per regime; ρ : Spearman coupling of the two percentile surfaces; HH: compounding population share at the median split.

City	CC	Region	Population	Grade	n_{em}	$\bar{D}_{ev}$	$\bar{C}_{em}$	G_{ev}	G_{em}	ρ	HH
Paris	FR	West	12,886,280	covered	2	0.135	0.73	0.625	0.497	0.50	0.334
Madrid	ES	South	7,601,394	covered	2	0.191	0.62	0.533	0.439	0.39	0.316
Barcelona	ES	South	5,945,709	covered	2	0.175	0.84	0.545	0.418	0.36	0.309
Milano	IT	South	5,030,024	covered	2	0.183	0.61	0.400	0.400	0.37	0.319
Berlin	DE	West	4,875,655	covered	2	0.158	0.62	0.558	0.438	0.38	0.317
Roma	IT	South	4,432,845	covered	2	0.287	1.14	0.424	0.503	0.48	0.342
Athina	EL	South	3,522,264	covered	1	0.103	0.73	0.609	0.500	0.28	0.274
Bruxelles	BE	West	3,521,497	covered	2	0.182	1.29	0.583	0.286	0.40	0.332
Napoli	IT	South	3,457,649	covered	2	0.326	0.83	0.305	0.384	0.15	0.270
Warszawa	PL	CEE	3,334,987	partial	2	0.225	1.99	0.589	0.590	0.49	0.347
Hamburg	DE	West	3,162,607	covered	2	0.260	0.58	0.543	0.437	0.40	0.321
Bucureşti	RO	CEE	3,140,502	desert	2	0.335	3.21	0.524	0.774	0.73	0.424
Lisboa	PT	South	3,132,904	covered	2	0.239	1.19	0.367	0.409	0.48	0.319
München	DE	West	3,054,507	covered	2	0.178	0.60	0.616	0.449	0.35	0.304
Budapest	HU	CEE	3,030,949	covered	2	0.183	1.05	0.538	0.507	0.48	0.327
Amsterdam	NL	West	2,829,893	covered	2	0.227	0.75	0.501	0.326	0.20	0.285
Stockholm	SE	North	2,549,795	partial	2	0.267	1.73	0.460	0.513	0.65	0.389
Köln	DE	West	2,438,338	covered	2	0.204	0.57	0.520	0.398	0.36	0.312
Praha	CZ	CEE	2,247,160	covered	2	0.282	1.11	0.515	0.475	0.46	0.354
København	DK	North	2,180,998	covered	2	0.254	0.94	0.401	0.401	0.30	0.307
Sofia	BG	CEE	1,647,936	partial	2	0.285	1.44	0.511	0.600	0.73	0.408
Helsinki	FI	North	1,642,879	covered	1	0.379	1.36	0.341	0.426	0.26	0.270
Oslo	NO	North	1,550,450	partial	2	0.209	2.06	0.553	0.467	0.39	0.297
Porto	PT	South	1,531,690	covered	2	0.324	0.85	0.272	0.395	0.42	0.321
Kraków	PL	CEE	1,350,286	covered	2	0.308	1.50	0.523	0.519	0.58	0.396
Katowice	PL	CEE	1,341,890	covered	2	0.168	1.02	0.489	0.279	0.10	0.267
Zagreb	HR	CEE	1,293,698	partial	2	0.343	2.07	0.499	0.583	0.70	0.395
Antwerpen	BE	West	1,219,366	covered	2	0.289	0.77	0.523	0.406	0.29	0.311
Rīga	LV	CEE	1,110,303	desert	2	0.362	3.37	0.502	0.682	0.55	0.374
Bilbao	ES	South	1,053,444	covered	2	0.160	1.06	0.570	0.433	0.42	0.324
Łódź	PL	CEE	1,045,886	covered	2	0.255	1.60	0.534	0.491	0.33	0.310
Göteborg	SE	North	1,042,720	partial	2	0.427	2.44	0.335	0.446	0.31	0.298
Vilnius	LT	CEE	1,010,376	desert	2	0.462	5.39	0.411	0.739	0.55	0.358
Palermo	IT	South	1,006,055	partial	2	0.317	3.64	0.383	0.292	0.27	0.289
Alicante	ES	South	996,867	covered	2	0.311	1.80	0.530	0.277	0.15	0.269
Wrocław	PL	CEE	939,076	covered	2	0.199	1.47	0.609	0.362	0.41	0.341
Bratislava	SK	CEE	931,613	covered	2	0.296	1.78	0.522	0.430	0.39	0.336
Ljubljana	SI	CEE	919,303	desert	2	0.447	4.96	0.387	0.481	0.48	0.341
Palma de Mallorca	ES	South	818,202	covered	2	0.251	1.39	0.503	0.453	0.14	0.294
Brno	CZ	CEE	745,397	covered	2	0.286	1.14	0.561	0.457	0.60	0.369
Tallinn	EE	CEE	731,883	desert	2	0.349	2.80	0.456	0.713	0.45	0.326
Grenoble	FR	West	688,170	covered	2	0.211	1.11	0.583	0.444	0.62	0.379
Luxembourg	LU	West	595,677	partial	2	0.312	1.64	0.469	0.407	0.30	0.301
Toulon	FR	West	540,279	covered	2	0.315	0.91	0.431	0.458	0.41	0.326
Aarhus	DK	North	525,545	covered	2	0.336	1.81	0.394	0.438	0.38	0.310
Chemnitz	DE	West	481,607	covered	2	0.223	0.69	0.558	0.363	0.36	0.319
Turku	FI	North	353,992	partial	2	0.409	1.63	0.385	0.565	0.67	0.360
Stavanger	NO	North	331,780	partial	2	0.252	2.10	0.498	0.492	0.28	0.299
Aalborg	DK	North	309,890	partial	2	0.405	1.87	0.419	0.541	0.47	0.342
Caserta	IT	South	298,402	covered	2	0.283	1.33	0.298	0.228	0.18	0.272
Vicenza	IT	South	280,054	covered	2	0.275	1.08	0.391	0.411	0.47	0.344
Helsingborg	SE	North	253,346	covered	2	0.420	0.90	0.284	0.387	0.51	0.366

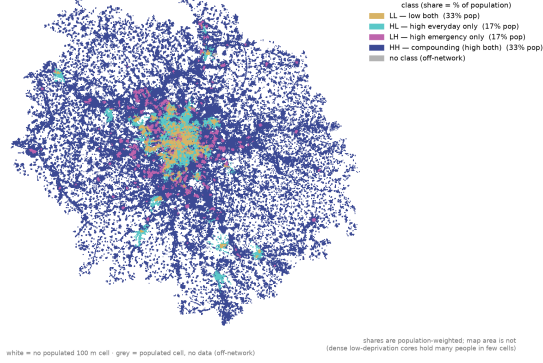
Table S2 continued

City	CC	Region	Population	Grade	n_{em}	$\bar{D}_{\text{ev}}$	$\bar{C}_{\text{em}}$	G_{ev}	G_{em}	ρ	HH
Jaén	ES	South	249,477	covered	2	0.291	1.24	0.543	0.385	-0.02	0.233
Szeged	HU	CEE	246,224	covered	1	0.281	0.82	0.533	0.568	0.58	0.392
Apeldoorn	NL	West	242,638	covered	2	0.326	1.02	0.439	0.341	0.33	0.333
Cosenza	IT	South	239,305	covered	2	0.505	1.23	0.407	0.500	0.70	0.390
Brăila	RO	CEE	235,919	partial	1	0.422	1.29	0.434	0.692	0.63	0.359
Amersfoort	NL	West	234,650	covered	2	0.219	0.64	0.458	0.350	0.26	0.282
Žilina	SK	CEE	217,546	covered	1	0.431	1.46	0.424	0.443	0.62	0.365
Västerås	SE	North	193,510	covered	2	0.454	1.13	0.327	0.478	0.62	0.387
Norrköping	SE	North	193,317	partial	1	0.445	1.88	0.423	0.560	0.55	0.353
Lahti	FI	North	191,532	covered	1	0.495	2.14	0.353	0.444	0.50	0.345
Landshut	DE	West	168,617	covered	2	0.364	0.68	0.492	0.438	0.49	0.361
Sittard-Geleen	NL	West	165,544	covered	2	0.440	0.83	0.316	0.273	0.31	0.287
Arezzo	IT	South	146,788	covered	2	0.381	1.02	0.412	0.446	0.45	0.342
Łomża	PL	CEE	117,255	partial	2	0.484	1.01	0.417	0.621	0.74	0.430
Talavera de la Reina	ES	South	101,050	covered	1	0.378	0.89	0.522	0.454	0.66	0.415

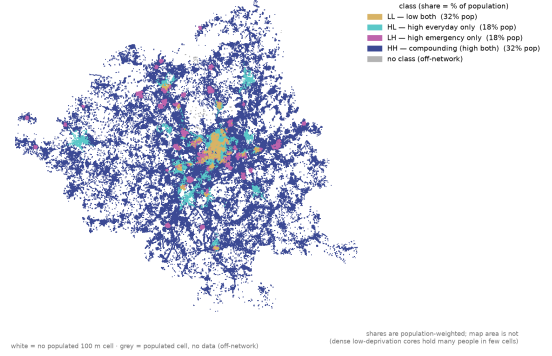
S6 Per-city compounding maps

The median-split co-location typology map of each of the 67 cities, ordered by population. Legends carry population-weighted class shares; the maps are area-weighted, so coloured area and quoted share will not match by eye. Read for spatial pattern only: roughly 24% of population flips class between routing engines (E.1), so no single cell's class is a finding.

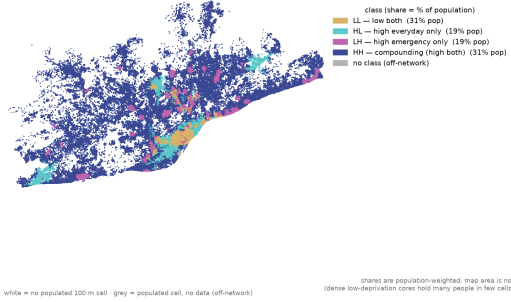
Paris: everyday x emergency co-location (pop-weighted median split of percentiles)



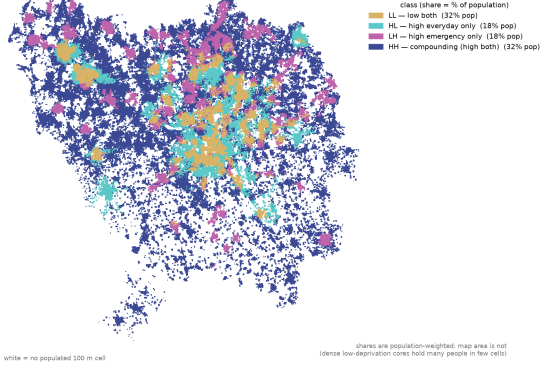
Madrid: everyday x emergency co-location (pop-weighted median split of percentiles)



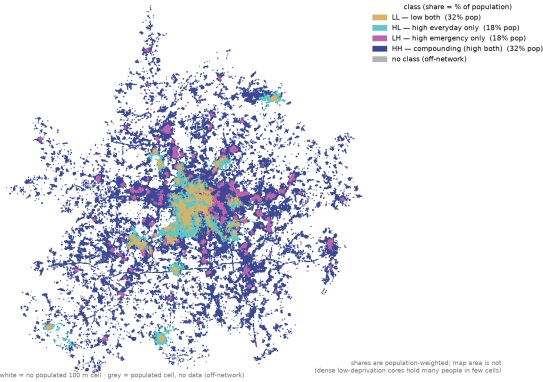
Barcelona: everyday x emergency co-location (pop-weighted median split of percentiles)



Milano: everyday x emergency co-location (pop-weighted median split of percentiles)



Berlin: everyday x emergency co-location (pop-weighted median split of percentiles)



Roma: everyday x emergency co-location (pop-weighted median split of percentiles)

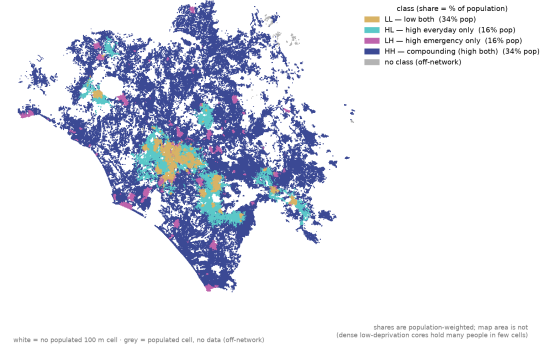
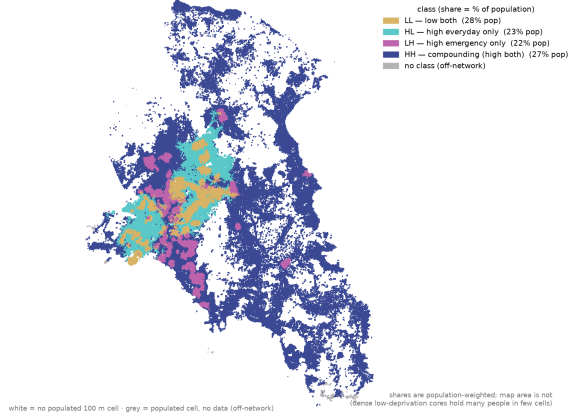
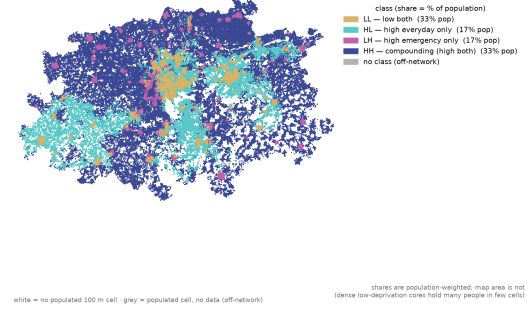


Figure S3: Per-city compounding maps (median-split co-location typology), cities 1–6 by population: Paris (FR), Madrid (ES), Barcelona (ES), Milano (IT), Berlin (DE), Roma (IT). Read for spatial pattern only; per-cell classes carry a ~24 % routing-engine flip share (E.1).

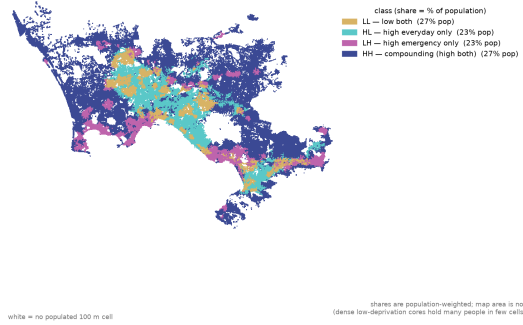
Athina: everyday x emergency co-location (pop-weighted median split of percentiles)



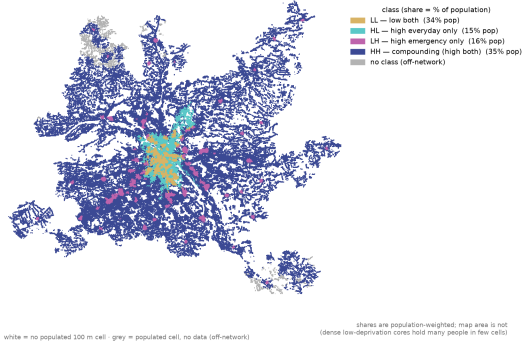
Bruxelles: everyday x emergency co-location (pop-weighted median split of percentiles)



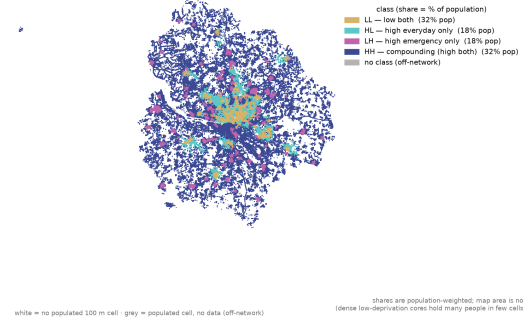
Napoli: everyday x emergency co-location (pop-weighted median split of percentiles)



Warszawa: everyday x emergency co-location (pop-weighted median split of percentiles)



Hamburg: everyday x emergency co-location (pop-weighted median split of percentiles)



București: everyday x emergency co-location (pop-weighted median split of percentiles)

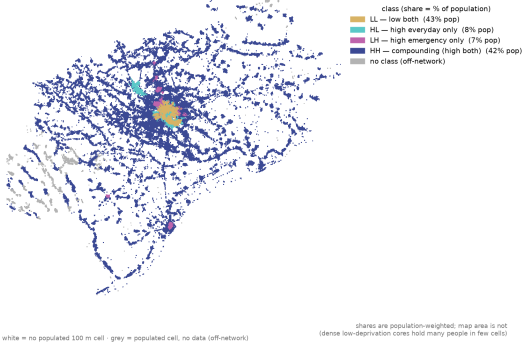
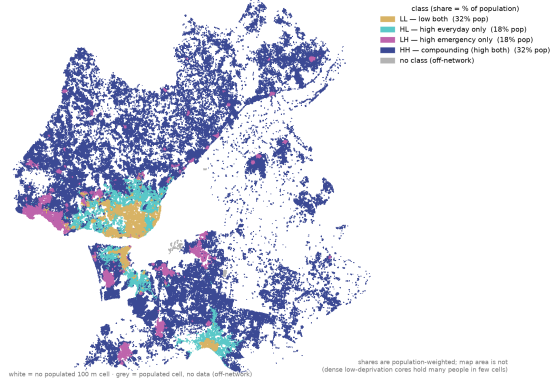
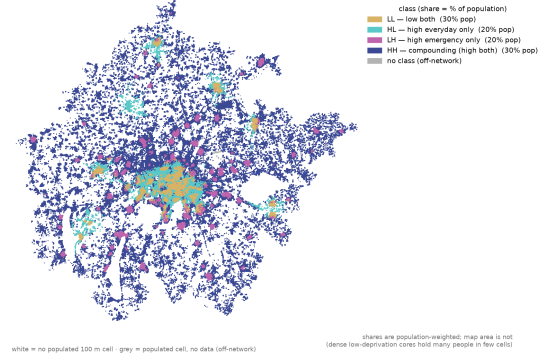


Figure S4: Per-city compounding maps (median-split co-location typology), cities 7–12 by population: Athina (EL), Bruxelles (BE), Napoli (IT), Warszawa (PL), Hamburg (DE), București (RO). Read for spatial pattern only; per-cell classes carry a $\sim 24\%$ routing-engine flip share (E.1).

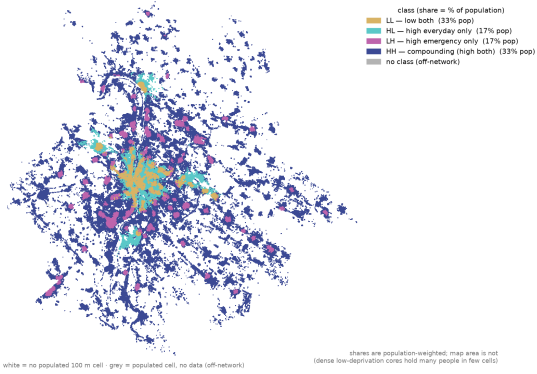
Lisboa: everyday x emergency co-location (pop-weighted median split of percentiles)



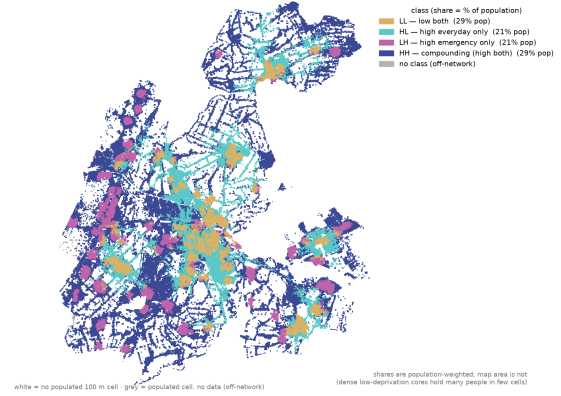
München: everyday x emergency co-location (pop-weighted median split of percentiles)



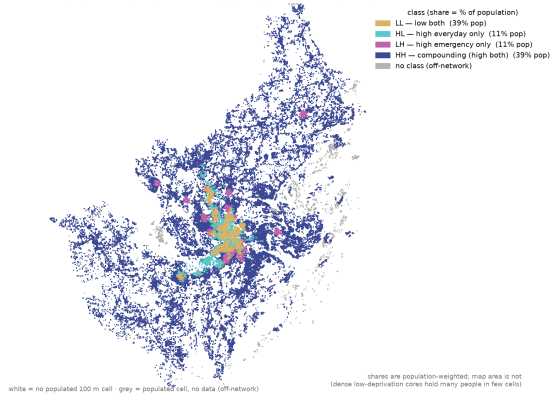
Budapest: everyday x emergency co-location (pop-weighted median split of percentiles)



Amsterdam: everyday x emergency co-location (pop-weighted median split of percentiles)



Stockholm: everyday x emergency co-location (pop-weighted median split of percentiles)



Köln: everyday x emergency co-location (pop-weighted median split of percentiles)

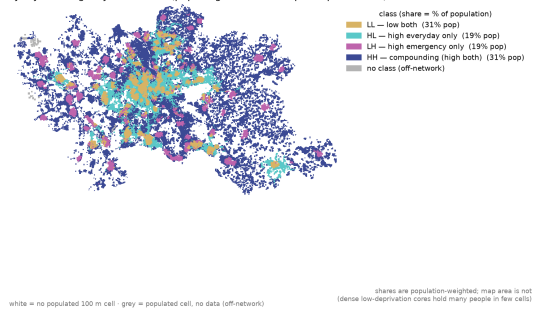


Figure S5: Per-city compounding maps (median-split co-location typology), cities 13–18 by population: Lisboa (PT), München (DE), Budapest (HU), Amsterdam (NL), Stockholm (SE), Köln (DE). Read for spatial pattern only; per-cell classes carry a ~24 % routing-engine flip share (E.1).

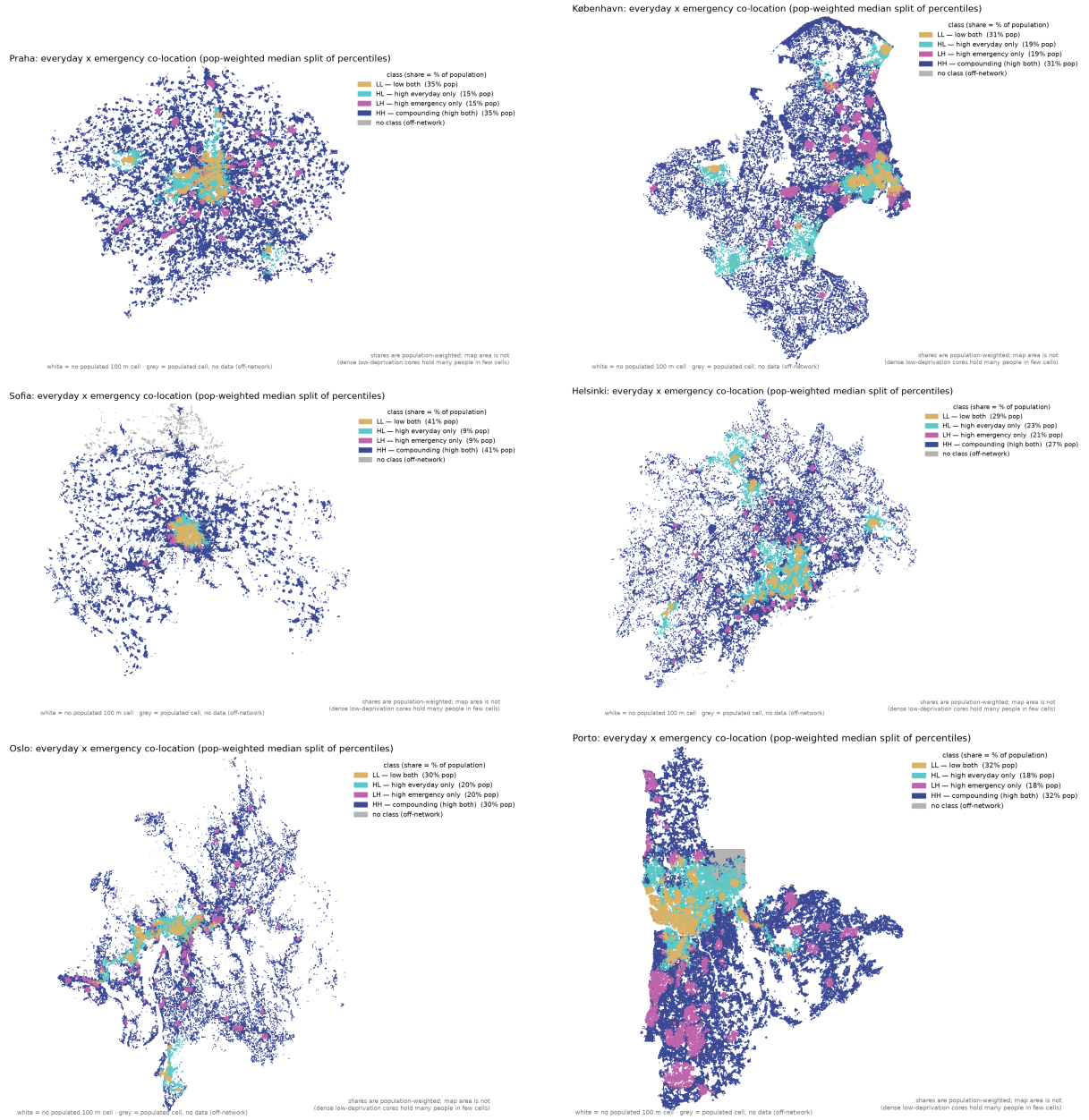


Figure S6: Per-city compounding maps (median-split co-location typology), cities 19–24 by population: Praha (CZ), København (DK), Sofia (BG), Helsinki (FI), Oslo (NO), Porto (PT). Read for spatial pattern only; per-cell classes carry a $\sim 24\%$ routing-engine flip share (E.1).

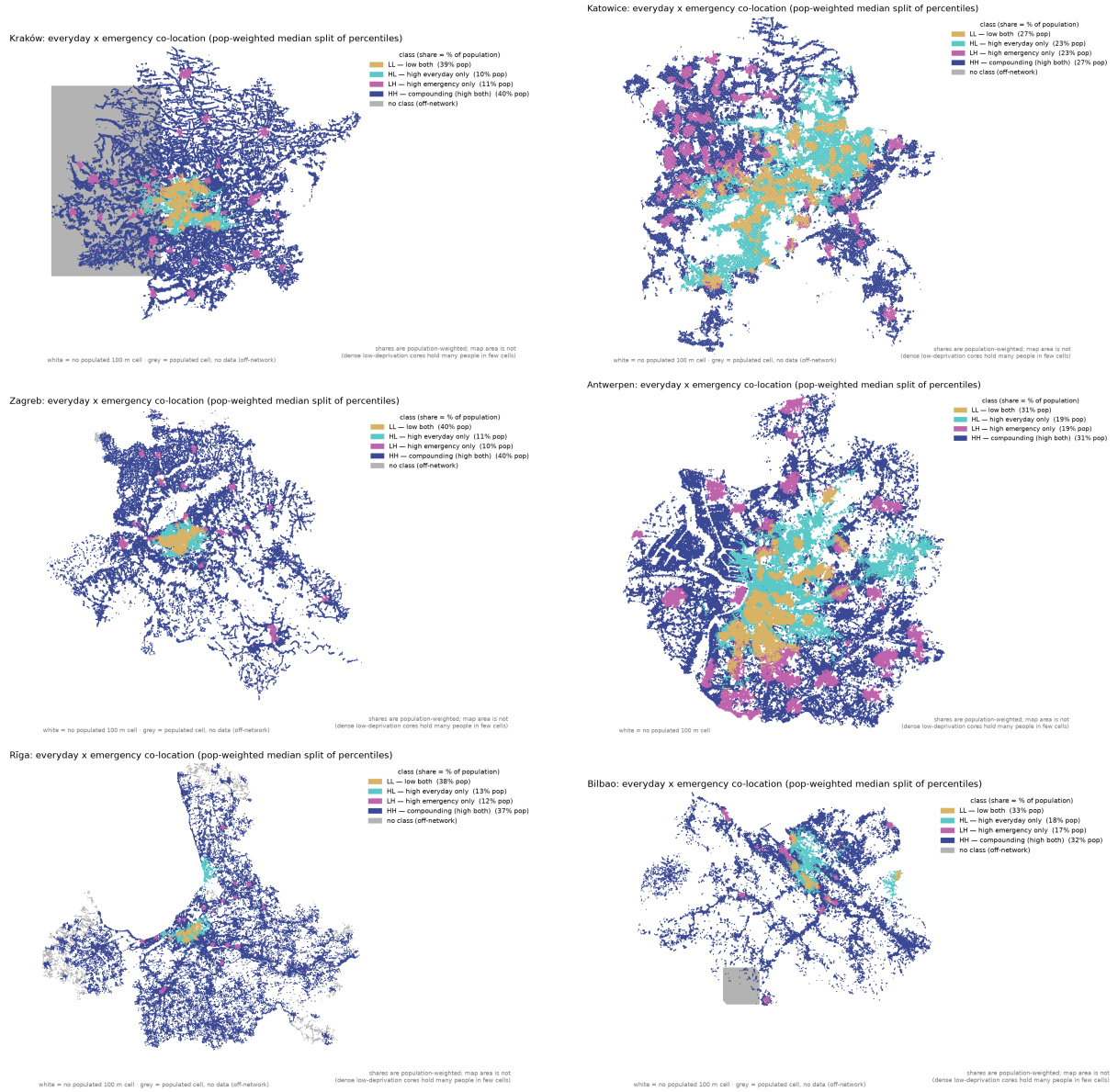


Figure S7: Per-city compounding maps (median-split co-location typology), cities 25–30 by population: Kraków (PL), Katowice (PL), Zagreb (HR), Antwerpen (BE), Rīga (LV), Bilbao (ES). Read for spatial pattern only; per-cell classes carry a ~24 % routing-engine flip share (E.1).

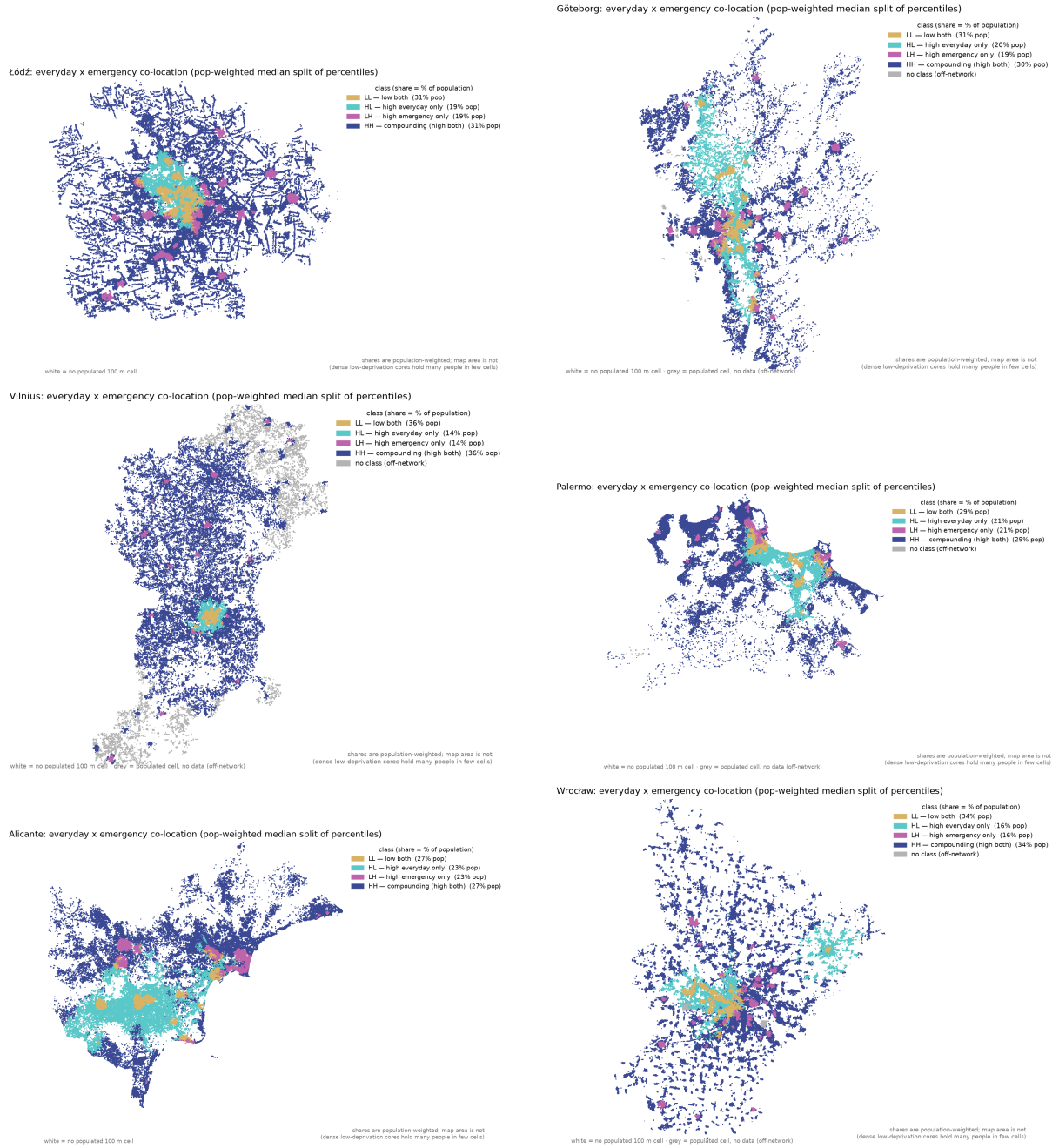
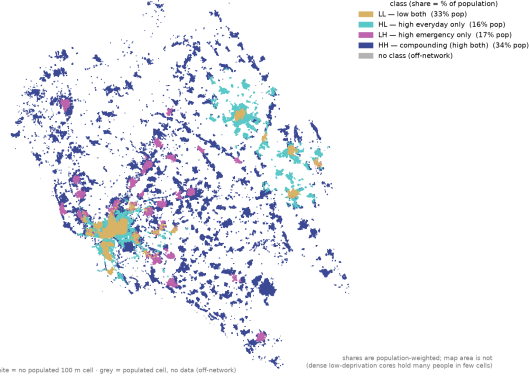
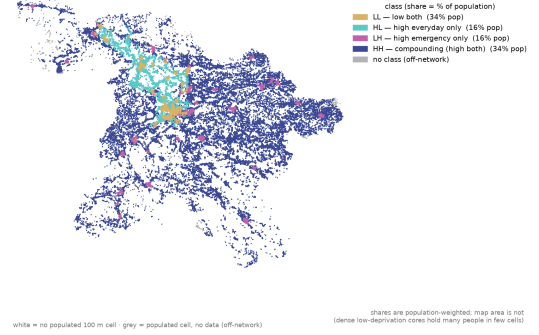


Figure S8: Per-city compounding maps (median-split co-location typology), cities 31–36 by population: Łódź (PL), Göteborg (SE), Vilnius (LT), Palermo (IT), Alicante (ES), Wrocław (PL). Read for spatial pattern only; per-cell classes carry a ~24 % routing-engine flip share (E.1).

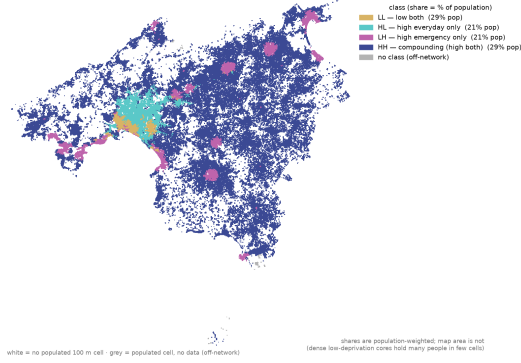
Bratislava: everyday x emergency co-location (pop-weighted median split of percentiles)



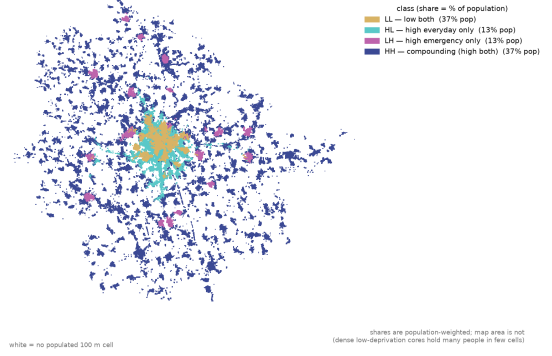
Ljubljana: everyday x emergency co-location (pop-weighted median split of percentiles)



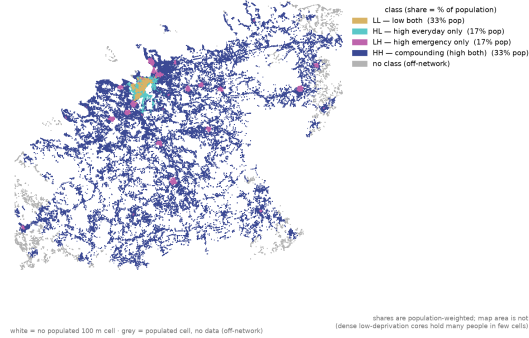
Palma de Mallorca: everyday x emergency co-location (pop-weighted median split of percentiles)



Brno: everyday x emergency co-location (pop-weighted median split of percentiles)



Tallinn: everyday x emergency co-location (pop-weighted median split of percentiles)



Grenoble: everyday x emergency co-location (pop-weighted median split of percentiles)

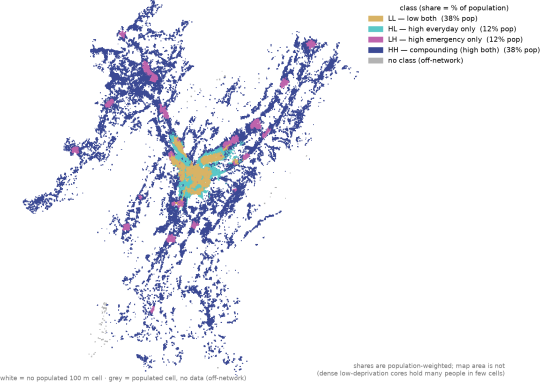
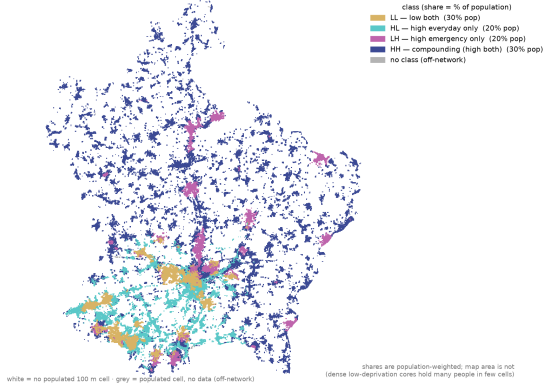
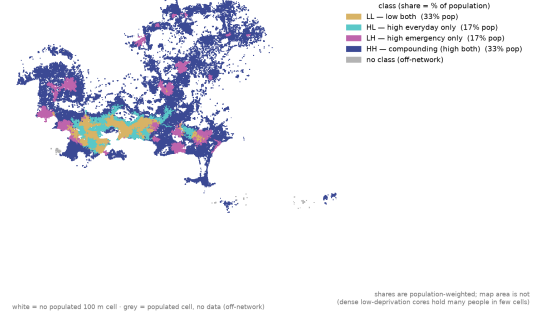


Figure S9: Per-city compounding maps (median-split co-location typology), cities 37–42 by population: Bratislava (SK), Ljubljana (SI), Palma de Mallorca (ES), Brno (CZ), Tallinn (EE), Grenoble (FR). Read for spatial pattern only; per-cell classes carry a ~24 % routing-engine flip share (E.1).

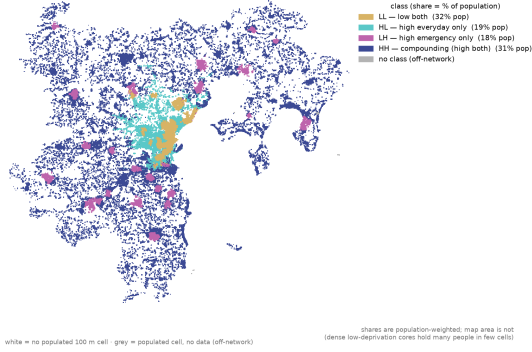
Luxembourg: everyday x emergency co-location (pop-weighted median split of percentiles)



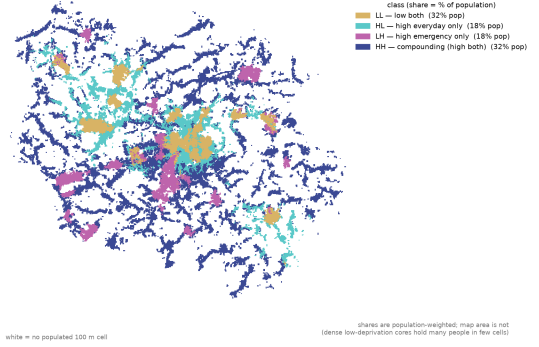
Toulon: everyday x emergency co-location (pop-weighted median split of percentiles)



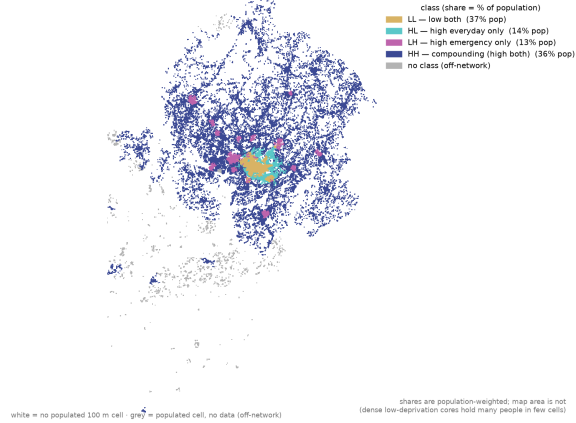
Aarhus: everyday x emergency co-location (pop-weighted median split of percentiles)



Chemnitz: everyday x emergency co-location (pop-weighted median split of percentiles)



Turku: everyday x emergency co-location (pop-weighted median split of percentiles)



Stavanger: everyday x emergency co-location (pop-weighted median split of percentiles)

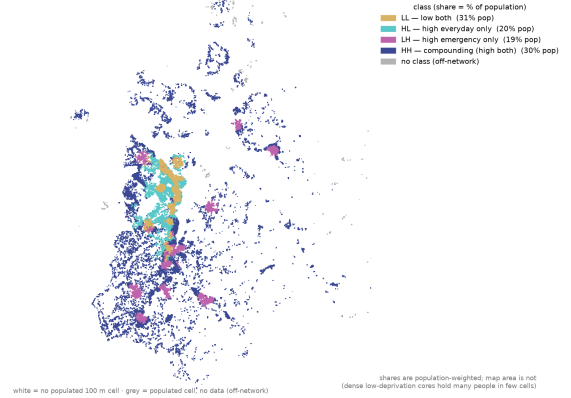


Figure S10: Per-city compounding maps (median-split co-location typology), cities 43–48 by population: Luxembourg (LU), Toulon (FR), Aarhus (DK), Chemnitz (DE), Turku (FI), Stavanger (NO). Read for spatial pattern only; per-cell classes carry a ~24 % routing-engine flip share (E.1).

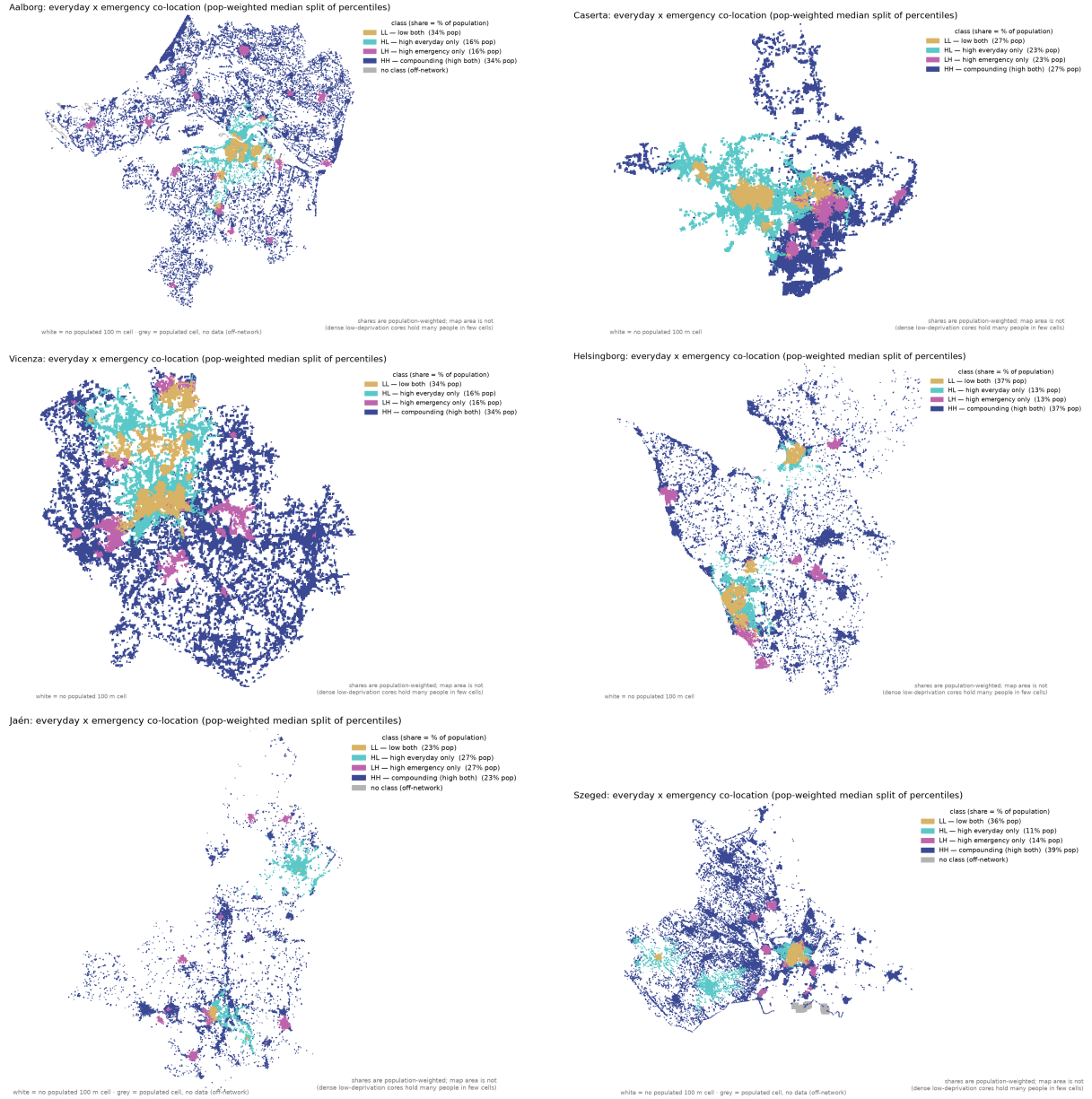


Figure S11: Per-city compounding maps (median-split co-location typology), cities 49–54 by population: Aalborg (DK), Caserta (IT), Vicenza (IT), Helsingborg (SE), Jaén (ES), Szeged (HU). Read for spatial pattern only; per-cell classes carry a $\sim 24\%$ routing-engine flip share (E.1).

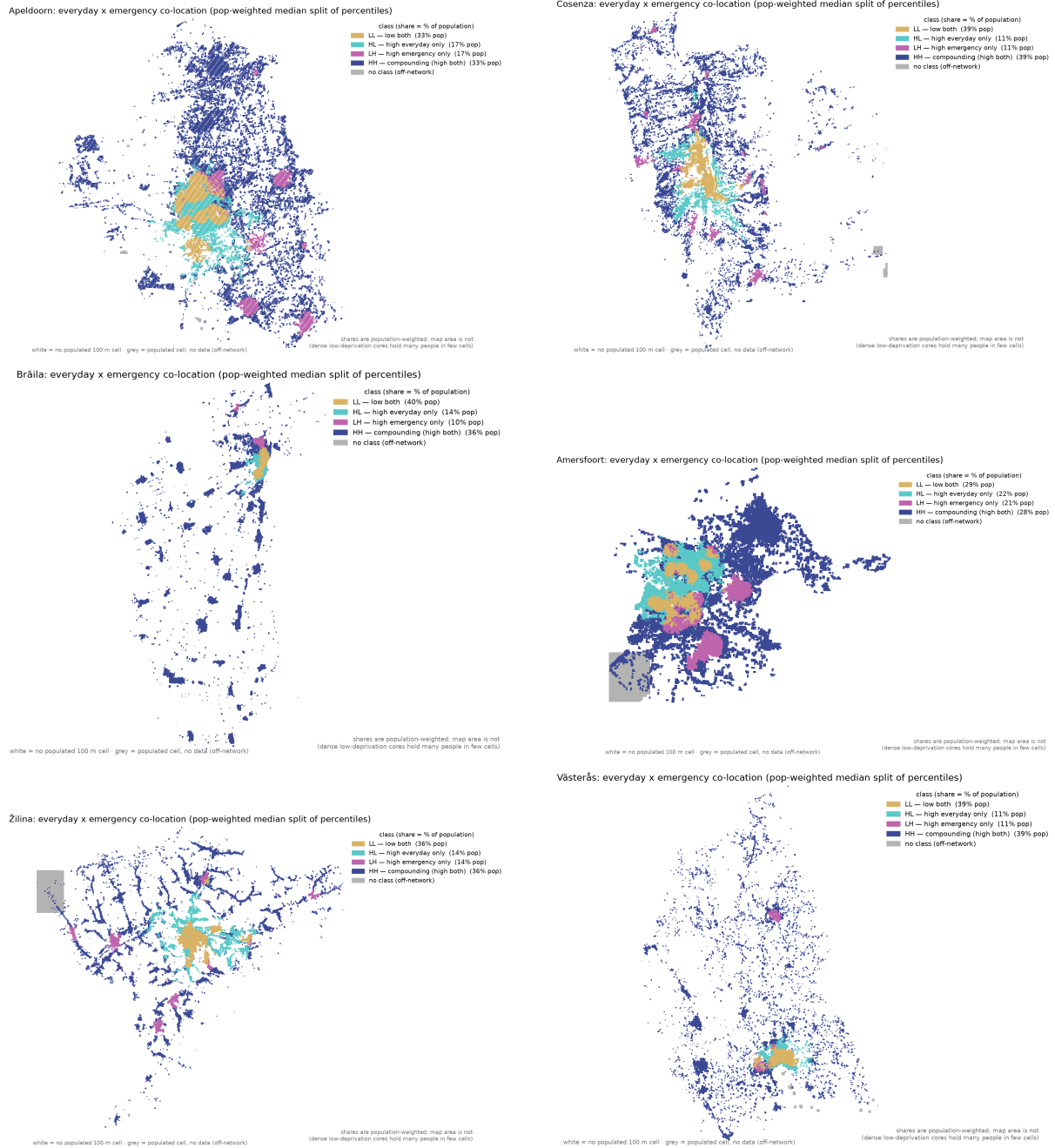


Figure S12: Per-city compounding maps (median-split co-location typology), cities 55–60 by population: Apeldoorn (NL), Cosenza (IT), Brăila (RO), Amersfoort (NL), Žilina (SK), Västerås (SE). Read for spatial pattern only; per-cell classes carry a $\sim 24\%$ routing-engine flip share (E.1).

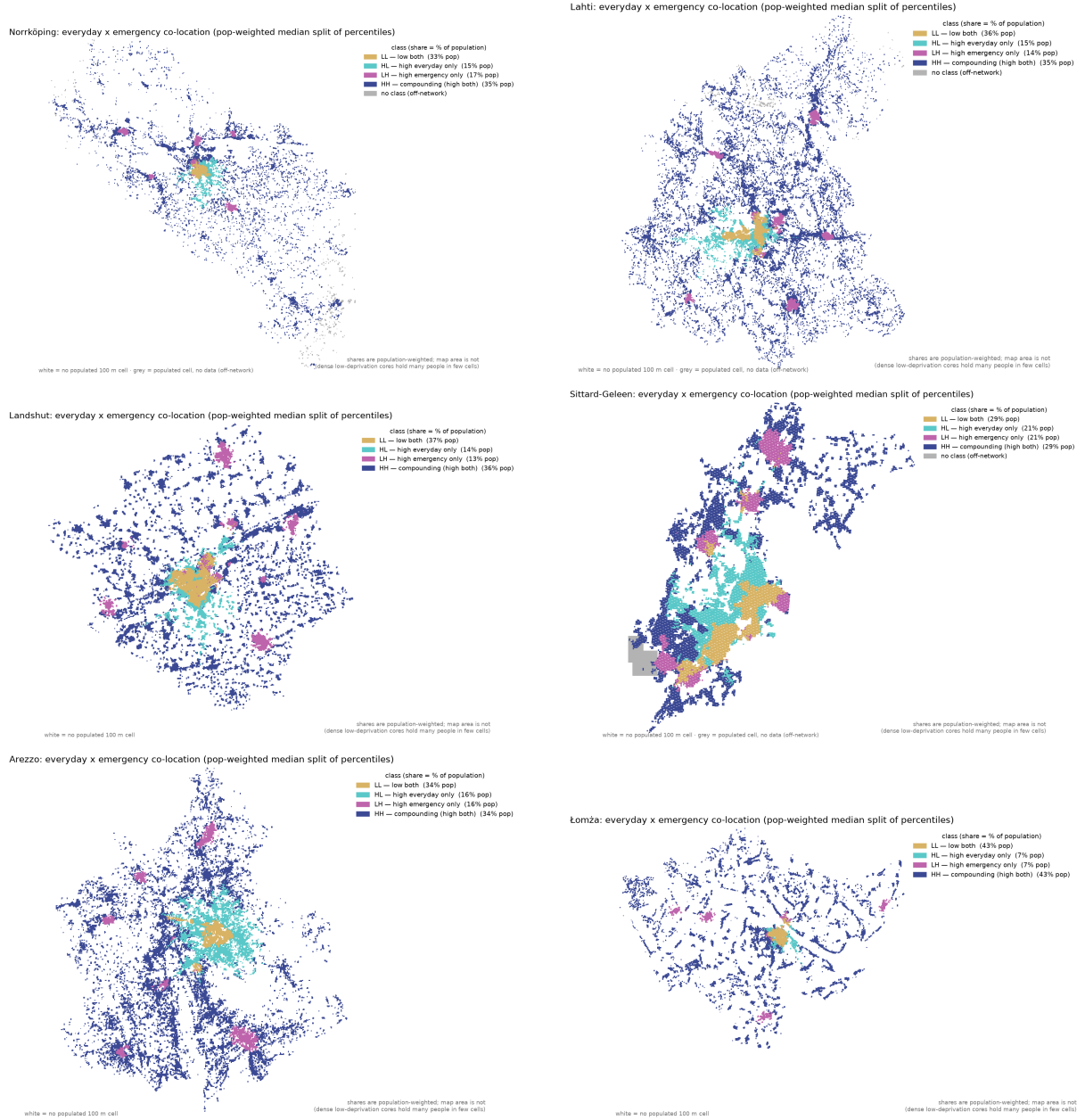


Figure S13: Per-city compounding maps (median-split co-location typology), cities 61–66 by population: Norrköping (SE), Lahti (FI), Landshut (DE), Sittard-Geleen (NL), Arezzo (IT), Łomża (PL). Read for spatial pattern only; per-cell classes carry a $\sim 24\%$ routing-engine flip share (E.1).

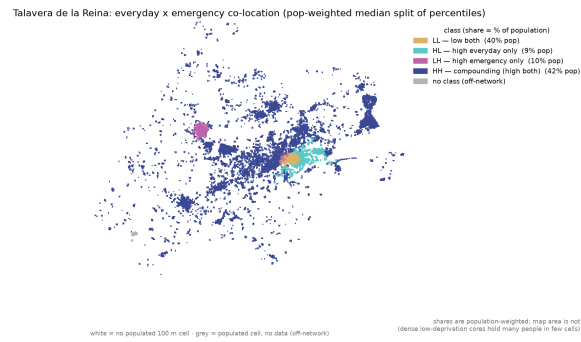


Figure S14: Per-city compounding maps (median-split co-location typology), cities 67–67 by population: Talavera de la Reina (ES). Read for spatial pattern only; per-cell classes carry a $\sim 24\%$ routing-engine flip share (E.1).